TUM School of Computation, Information and Technology

Department of Computer Science (Data Analytics and Machine Learning)

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Mathematics in Science and Engineering

Uncertainty Quantification for Machine Learning Models in Transportation Policy Analysis

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Unsicherheitsquantifizierung für maschinelle Lernmodelle in der Verkehrspolitikanalyse

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I confirm that this master's thesis is my own work and I have documented all sources and material used.

Munich, May 15, 2026 Mohd Zamin Quadri

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Abstract

Agent-based simulators such as MATSim are the standard tool for evaluating urban transport policy but take hours per Paris-scale scenario; Graph Neural Network (GNN) surrogates close that gap to seconds. Accuracy alone is not enough for decision support, however, because predictions can degrade outside the training distribution without any external signal. This thesis evaluates uncertainty quantification (UQ) for a PointNet-TransfGAT GNN traffic surrogate on a Paris road network, using a fixed 1,000-scenario subset of the 10,000-scenario MATSim corpus reused across all eleven trials. The scope is intentionally narrow: one road network, one intervention type (capacity reduction), one GNN architecture family, one subset. The findings are specific to that setting, not general claims.

Trial 8 ($R^2 = 0.5957$ on 3,163,500 test nodes) is the primary UQ base. Monte Carlo Dropout gives Spearman $\rho = 0.4820$ between σ and absolute error, but the raw intervals are uncalibrated ($k_{95} = 11.66$ vs Gaussian 1.96); regression σ-scaling at $T = 2.887$ reduces the Kuleshov ECE by 90.5% and brings 1σ coverage to within 0.3 pp of the Gaussian target. Split conformal achieves marginal $\text{PICP}_{90} = 90.02\%$ and marginal $\text{PICP}_{95} = 95.01\%$; adaptive conformal narrows conditional coverage from $[59.0\%, 98.1\%]$ to $[83.7\%, 96.4\%]$ across uncertainty deciles. Retaining the 50% most confident predictions reduces MAE by 41.2% to 2.32 veh/h.

Three uncertainty-aware training extensions isolate one design choice. Trial 10 (CQR head, unfrozen backbone) collapses to $R^2 = 0.4057$, while Trial 11 (same head and learning rate, backbone frozen) passes all six gates at $R^2 = 0.5835$; the contrast supports freezing the MSE-trained backbone in this setup and stops short of any broader claim. A five-member Deep Ensemble reaches the highest GNN-based, UQ-capable point accuracy ($R^2 = 0.6841$, $+14.8\%$ over T8) at a weaker UQ ranking ($\rho = 0.3997$). Non-GNN baselines (XGBoost $R^2 = 0.7414$, Random Forest 0.6612) exceed T8 on point accuracy. The main contribution is not point accuracy but a per-prediction UQ pipeline around the GNN surrogate (MC Dropout, regression σ-scaling, conformal coverage, frozen-backbone CQR) that the default tree baselines do not provide; tree-native UQ extensions such as quantile regression forests and NGBoost are left for follow-up work.

Zusammenfassung

Agentenbasierte Simulationen wie MATSim sind das Standardwerkzeug für die Bewertung städtischer Verkehrspolitik, benötigen jedoch Stunden pro Paris-Szenario; Graph-Neural-Network-Surrogate (GNN) reduzieren diese Laufzeit auf Sekunden. Genauigkeit allein genügt jedoch nicht für Entscheidungsunterstützung: Vorhersagen können sich außerhalb der Trainingsverteilung unbemerkt verschlechtern. Diese Arbeit untersucht Unsicherheitsquantifizierung (UQ) für ein PointNetTransfGAT-GNN-Verkehrssurrogat am Pariser Straßennetz auf Basis einer festen Teilmenge von 1 000 aus 10 000 MATSim-Szenarien, die in allen elf Versuchen identisch verwendet wird.

Versuch 8 ($R^2 = 0{,}5957$ auf 3 163 500 Testknoten) ist das primäre UQ-Basismodell. Monte-Carlo-Dropout liefert Spearman $\rho = 0{,}4820$ zwischen σ und absolutem Fehler, doch die rohen Intervalle sind unkalibriert ($k_{95} = 11{,}66$ vs Gauß 1,96); eine Regressions-σ-Skalierung mit $T = 2{,}887$ reduziert den Kuleshov-ECE um 90,5 % und bringt die 1σ-Abdeckung innerhalb von 0,3 Prozentpunkten an das Gauß-Ziel. Split-Konformalprädiktion erreicht marginal $\text{PICP}_{90} = 90{,}02\,\%$ und marginal $\text{PICP}_{95} = 95{,}01\,\%$; adaptive Konformalprädiktion verengt die bedingte Abdeckung von [59,0 %, 98,1 %] auf [83,7 %, 96,4 %] über die σ-Dezile. Das Behalten der 50 % zuverlässigsten Vorhersagen reduziert den MAE um 41,2 % auf 2,32 Fz/h.

Drei unsicherheitsbewusste Trainingsvarianten isolieren einen Designaspekt: Versuch 10 (CQR-Kopf, trainierbarer Backbone) bricht auf $R^2 = 0{,}4057$ ein, während Versuch 11 (gleicher Kopf und gleiche Lernrate, eingefrorener Backbone) alle sechs Kriterien bei $R^2 = 0{,}5835$ besteht – der Vergleich stützt das Einfrieren des MSE-trainierten Backbones im hier untersuchten Aufbau und macht keinen darüber hinausgehenden Anspruch. Ein Deep Ensemble aus fünf Modellen erreicht die höchste GNN-basierte, UQ-fähige Punktgenauigkeit ($R^2 = 0{,}6841$, +14,8 % über T8) bei schwächerem UQ-Ranking ($\rho = 0{,}3997$). Nicht-GNN-Vergleichsmodelle (XGBoost $R^2 = 0{,}7414$, Random Forest 0,6612) übertreffen T8 in der Punktgenauigkeit. Der zentrale Beitrag liegt nicht in der Punktgenauigkeit, sondern in einer vorhersagespezifischen UQ-Pipeline um das GNN-Surrogat (MC-Dropout, Regressions-σ-Skalierung, Konformalprädiktion, CQR mit eingefrorenem Backbone), die die Baum-Vergleichsmodelle in ihrer Standardform nicht bereitstellen; baumbasierte UQ-Erweiterungen wie Quantile Regression Forests und NGBoost verbleiben als zukünftige Arbeit.

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1. Introduction

Urban traffic is hard to describe analytically. Millions of daily trips produce congestion patterns that emerge from the interaction of many individual choices, and a policy intervention – closing an arterial, reducing road capacity, opening a new metro line – reshapes those patterns through the responses of individual travellers. Agent-based simulators such as MATSim handle this kind of question by modelling each traveller's daily activity, mode and route choices, then iterating until the system converges to an approximate user equilibrium [1–3]. The price of that fidelity is computational: a single Paris-scale scenario takes more than eight hours on modern hardware [4]. For a planner who wants to compare a dozen variations of a road-pricing scheme, that cost is prohibitive.

Machine-learning surrogates promise to compress the loop from hours to seconds. Natterer, Rao, Tejada Lapuerta, et al. [4] showed that Graph Neural Networks (GNNs) are particularly well suited to this task: they exploit the topology of the road network natively while learning per-segment volume responses to capacity changes. Their PointNetTransfGAT architecture is the model the present work builds on.

Accuracy alone is not enough for decision support. A model trained on a limited set of scenarios will eventually face inputs unlike anything it has seen, and at that point its predictions can deteriorate without any external indication that something has changed. Without uncertainty quantification (UQ), planners cannot separate the reliable predictions from the unreliable ones, and a confident-but-wrong output can drive a poor infrastructure decision. This thesis addresses that gap by evaluating Monte Carlo Dropout, deep ensembles, conformal prediction, post-hoc regression σ-scaling, and several uncertainty-aware training variants on a GNN surrogate for policy-induced changes in traffic volume. The empirical work is deliberately constrained to one road network (Paris), one intervention type (capacity reduction), one GNN architecture family (PointNetTransfGAT), and one fixed 1,000-scenario subset; Section 6.10 enumerates the threats to validity that bound the resulting claims.

A note on scale. Trial 8's absolute test $R^2 \approx 0.60$ on the 1,000-scenario subset is below the $R^2 > 0.9$ reported by Natterer, Rao, Tejada Lapuerta, et al. [4] for the same architecture at the full 10,000-scenario scale. The contribution here is the *relative* ranking and calibration behaviour of UQ methods, which is what method-to-method comparisons on a fixed subset can establish; absolute point-accuracy at this scale is not claimed to be deployment-grade, and replicating the UQ comparisons at full scale is the most direct next step (Section 6.9).

Research questions. The investigation is organised around five research questions.

RQ1. How effectively does MC Dropout rank uncertainty for a GNN traffic surrogate?

RQ2. How does MC Dropout compare with ensemble-based UQ?

RQ3. Does combining MC Dropout with seed-ensemble variance add benefit?

RQ4. Can post-hoc calibration and conformal methods deliver reliable intervals without retraining the backbone?

RQ5. Can uncertainty-aware training extensions improve native UQ while preserving accuracy?

Each is answered, with supporting evidence and the relevant limitation, in Section 6.6.

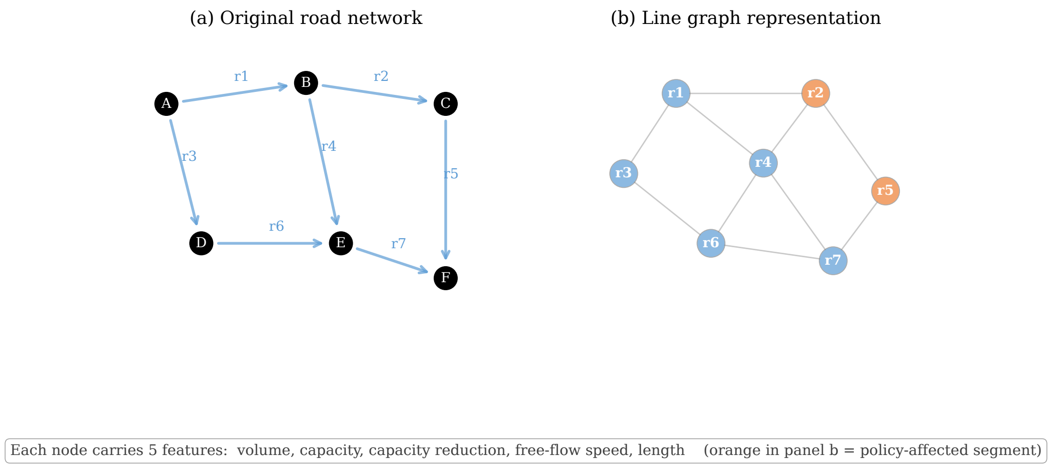

Figure 1.1.: The Paris road network represented as a directed line graph. Each road segment is a node, and edges connect adjacent segments. Each node carries five input features describing the segment's traffic state and physical properties. The full Paris network contains 31,635 nodes; with 100 graphs in the test set, this gives 3,163,500 individual node-level test predictions. Adapted from Natterer, Rao, Tejada Lapuerta, et al. [4].

1.1. Contributions

1. **A systematic UQ evaluation for a GNN traffic surrogate.** Six methods (MC Dropout, seed-ensemble variance, multi-model ensembles, deep ensembles, heteroscedastic regression, CQR) compared on a Paris road-network dataset with 3,163,500 node-level test predictions across 100 scenarios.

2. **An empirical characterisation of MC Dropout for GNN traffic surrogates.** For Trial 8, the pooled (node-level) Spearman $\rho = 0.4820$ at $S = 30$ samples; the mean per-graph ρ is 0.464 with bootstrap 95% CI $[0.460, 0.469]$. A formal S-convergence study confirms that ρ has converged by $S = 30$ (the gain from $S = 30$ to $S = 50$ is only $+1.03\%$).

3. **Practical limits of single-architecture ensembles.** Five seeded MC Dropout runs raise ρ to 0.4908; seed ensemble variance gives $\rho = 0.4370$. A weighted multi-model ensemble of five trials reaches $\rho = 0.4333$ and $R^2 = 0.5656$, below the best single model.

4. **A layered post-hoc calibration framework.** Split conformal achieves marginal $\text{PICP}_{90} = 90.02\%$ and marginal $\text{PICP}_{95} = 95.01\%$. Regression σ-scaling with $T = 2.887$ reduces ECE from 0.356 to 0.034 (-90.5%) and brings 1σ coverage from 32.7% to 68.0%. Adaptive conformal narrows conditional coverage from $[59.0\%, 98.1\%]$ to $[83.7\%, 96.4\%]$.

5. **Selective prediction and error detection.** Retaining the most confident 50% of predictions reduces MAE by 41.2% to 2.32 veh/h; AUROC for top-10% error detection reaches 0.7548.

6. **An empirical observation about freezing the backbone.** When the same MSE-trained backbone is reused with three different uncertainty-aware heads, the result depends on whether the backbone is frozen. Trial 9 (heteroscedastic head, frozen backbone) is partial-positive at $k_{95} = 2.84$ and $R^2 = 0.4991$; Trial 10 (CQR, backbone unfrozen) is a negative result at $R^2 = 0.4057$ with three of six gates failing; Trial 11, which uses the same CQR head as Trial 10 but freezes the backbone, passes all six gates with $R^2 = 0.5835$, $\text{PICP}_{90} = 89.82\%$ and $\text{PICP}_{95} = 94.91\%$. The T10/T11 pair shares head, loss, data, calibration protocol, and learning rate (5×10^{-4}); their single design difference is the backbone-trainability flag. The observation is reported for the architecture, subset, and loss/head pairings tested here, not as a general claim.

7. **A Deep Ensemble baseline.** Five independently seeded models reach $R^2 = 0.6841$ ($+14.8\%$ over T8) but a weaker UQ ranking ($\rho = 0.3997$, $k_{95} = 15.18$). The

Deep Ensemble is the strongest GNN-based point predictor evaluated, while MC Dropout produces the stronger uncertainty ranking; this complementarity is discussed in Chapter 6. XGBoost on the same five node features (Section 5.11) exceeds the Deep Ensemble on point accuracy at $R^2 = 0.7414$.

1.2. Thesis Structure

The five research questions stated above guide the chapter sequence. Chapter 2 surveys the theoretical foundations. Chapter 3 presents the architecture and the UQ methods used here. Chapter 4 covers the dataset and the experimental design. Chapter 5 reports the empirical findings and closes with an illustrative decision framework. Chapter 6 answers the research questions, summarises the primary findings, discusses the freezing-backbone observation, and lists the limitations, threats to validity and future work. Chapter 7 offers closing remarks, and Appendix A collects the verified master tables.

2. Background and Related Work

2.1. Agent-Based Transport Models

Rather than aggregating behaviour into mean-field equations, agent-based models (ABMs) assign individual decision logic to each entity in the system, with macroscopic patterns produced by the cumulative interactions of those entities [2, 3]. In transport applications, agents stand for individual travellers who decide on activities, departure times, modes, and routes; macroscopic patterns then emerge from the interactions on a shared infrastructure.

The Multi-Agent Transport Simulation (MATSim) [1] is the open-source platform used here. Within MATSim, each synthetic traveller is initialised with a daily plan covering activities, modes, and routes; a queue-based mesoscopic loader executes those plans on the network, and a co-evolutionary replanning loop iterates until a Nash-style equilibrium is reached. Simulating the Paris metropolitan region requires more than eight hours per scenario on modern hardware [4], the cost that motivates surrogate modelling.

2.2. Graph Neural Networks

Graph Neural Networks (GNNs) extend deep learning to graph-structured data [5–8] and follow the message-passing paradigm of Gilmer et al. [9]. A graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ has node features $\mathbf{X}$; the architectures used here do not consume edge-level features. Each node v has an incoming neighbourhood $\mathcal{N}(v) = \{u : (u, v) \in \mathcal{E}\}$, and messages are aggregated over that set. Starting from $\mathbf{h}_v^{(0)} = \mathbf{x}_v$, a message-passing network applies K rounds of aggregation and update:

$$\mathbf{m}_v^{(k)} = \mathrm{A}^{(k)}\left(\left\{\mathbf{h}_u^{(k-1)} : u \in \mathcal{N}(v)\right\}\right), \tag{2.1}$$

$$\mathbf{h}_v^{(k)} = \mathrm{U}^{(k)}\left(\mathbf{h}_v^{(k-1)},\ \mathbf{m}_v^{(k)}\right), \tag{2.2}$$

where the aggregation function is permutation-invariant. Graph Attention Networks (GATs [10]) replace fixed aggregation weights with learned attention coefficients, weighting neighbour contributions by relevance.

Traffic prediction with GNNs and the line-graph representation. Most prior GNN work on traffic targets a temporal task: given a window of past observations, predict the next time steps; intersections are graph nodes and road links are edges. This thesis takes the alternative line-graph view, in which each road segment becomes a node and edges encode segment adjacency. The line graph $L(G)$ is a classical graph-theoretic object; the practical motivation for using it here is that the prediction target, the change in volume induced by a policy intervention, is naturally per-segment. Following Natterer et al. [4], each node carries the policy-relevant features (volume, capacity, capacity reduction, free-flow speed, length) and the model produces a per-segment Δv. Geometric coordinates of each segment are preserved as auxiliary node features and consumed by the PointNetConv layers [11] of the architecture described in Chapter 3.

2.3. Uncertainty Quantification in Deep Learning

Uncertainty quantification (UQ) lets a model express how confident it is in its outputs [12, 13]. Two kinds are distinguished: *aleatoric* uncertainty reflects intrinsic stochasticity of the data-generating process and cannot be eliminated by collecting more data; *epistemic* uncertainty stems from gaps in training coverage and shrinks as the dataset grows or the model capacity is better matched to the task. Bayesian neural networks [14] formalise epistemic uncertainty by a posterior over weights, but exact inference is intractable for non-trivial networks.

2.3.1. Monte Carlo Dropout and Deep Ensembles

Gal and Ghahramani [15] showed that a network trained with dropout can be interpreted as a variational approximation to a Bayesian neural network. Keeping dropout active at inference time and averaging S stochastic forward passes,

$$\hat{y}_s = f_{\hat{\mathbf{w}}_s}(\mathbf{x}), \quad \mathbb{E}[y] \approx \frac{1}{S}\sum_s \hat{y}_s, \quad \mathrm{Var}[y] \approx \frac{1}{S}\sum_s (\hat{y}_s - \mathbb{E}[y])^2, \tag{2.3}$$

yields samples from the approximate predictive distribution. MC Dropout requires no architectural change at training time and can be applied post-hoc to any existing model that already includes dropout layers. Its variational posterior, however, is typically too concentrated, so the raw σ from MC Dropout is not a calibrated standard deviation in absolute terms [16], a property quantified empirically in Chapter 5 via the scaling factor k_{95}. Deep ensembles [17] replace stochastic dropout sampling with M independently trained networks; the ensemble mean is the prediction and the inter-member variance is the uncertainty, at the cost of $M\times$ training and inference compute.

2.3.2. Conformal Prediction

Conformal prediction [18, 19] is a distribution-free framework for finite-sample marginal coverage, requiring only the *exchangeability* of the data. Given a model f, a calibration set $\mathcal{D}_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^{n}$, and target level $1-\alpha$, split conformal computes nonconformity scores $s_i = |y_i - f(x_i)|$, sets $q = \text{Quantile}(s_1, \dots, s_n;\ \lceil (n+1)(1-\alpha) \rceil / n)$, and returns intervals $\mathcal{C}(x_{\text{new}}) = [f(x_{\text{new}}) - q,\ f(x_{\text{new}}) + q]$ with $\Pr(y_{\text{new}} \in \mathcal{C}) \geq 1-\alpha$.

When an auxiliary σ_i (e.g. from MC Dropout) is available, the score can be normalised, $s_i^{\text{adapt}} = |y_i - f(x_i)|/(\sigma_i + \varepsilon)$ for a small $\varepsilon > 0$, producing input-adaptive widths $[\hat{y} - q^{\text{adapt}}\sigma,\ \hat{y} + q^{\text{adapt}}\sigma]$ [20] while preserving the marginal guarantee. Barber et al. [21] prove that distribution-free *conditional* coverage is impossible without further assumptions; standard conformal can therefore over-cover low-uncertainty inputs and under-cover high-uncertainty ones, a pattern observed empirically in Chapter 5, while adaptive conformal partly mitigates this in practice.

2.3.3. Heteroscedastic Regression and the NLL–MSE Trade-off

A heteroscedastic head outputs a mean $\hat{\mu}$ and a log-variance $\log \hat{\sigma}^2$ per node, trained by minimising the Gaussian negative log-likelihood

$$\mathcal{L}_{\text{NLL}} = \frac{1}{n} \sum_{i=1}^{n} \left[\frac{(y_i - \hat{\mu}_i)^2}{2\hat{\sigma}_i^2} + \tfrac{1}{2} \log \hat{\sigma}_i^2 \right], \tag{2.4}$$

where the Gaussian normalising constant $\frac{1}{2}\log 2\pi$ is omitted. The loss can be reduced either by improving $\hat{\mu}$ or by inflating $\hat{\sigma}$ on hard examples. Seitzer, Tavakoli, Antic, and Martius [22] analyse the joint optimisation of $(\hat{\mu}, \hat{\sigma})$ and observe a degenerate regime in which the gradient pathway through $\hat{\sigma}$ dominates whenever the mean head has insufficient capacity, leaving $\hat{\mu}$ under-fit and $\hat{\sigma}$ inflated on hard examples. A regulariser $\lambda \sum_i (\log \hat{\sigma}_i^2)^2$ is commonly added to keep optimisation stable [22]. This trade-off is the central reason why Trial 9 here trades R^2 for a roughly four-fold improvement in k_{95} (from 11.66 to 2.84).

2.3.4. Conformalized Quantile Regression

Conformalized Quantile Regression (CQR) [20] obtains prediction intervals that are both adaptive and conformally calibrated. A model is trained to predict the lower and upper quantiles $\hat{q}^{\text{lo}}$ and $\hat{q}^{\text{hi}}$ via the joint pinball loss

$$\mathcal{L}_{\text{pinball}} = \frac{1}{n} \sum_{i=1}^{n} \left[\rho_{\tau_{\text{lo}}}(y_i - \hat{q}_i^{\text{lo}}) + \rho_{\tau_{\text{hi}}}(y_i - \hat{q}_i^{\text{hi}}) \right], \quad \rho_\tau(u) = u(\tau - \mathbf{1}[u < 0]). \tag{2.5}$$

A single conformal correction $\hat{Q}$ is then computed on a held-out set, and intervals become $[\hat{q}^{\text{lo}} - \hat{Q},\ \hat{q}^{\text{hi}} + \hat{Q}]$.

2.3.5. Selective Prediction and Evaluation Metrics

Selective prediction [23], originally formulated for classification but easily extended to regression, ranks predictions by an uncertainty score σ_i and retains only the most confident fraction τ, characterising how much an uncertainty signal can sharpen accuracy when an abstention budget is available.

Several complementary metrics evaluate uncertainty quality. The Spearman rank correlation ρ between σ and $|y - \hat{y}|$ measures whether large uncertainty tracks large error and is invariant to monotone rescaling. The Expected Calibration Error [24, 25] summarises how observed coverage deviates from nominal across the full $[0, 1]$ range. The Continuous Ranked Probability Score (CRPS) [26] is a proper scoring rule that for a Gaussian predictive distribution admits the closed form

$$\text{CRPS}(\mathcal{N}(\mu, \sigma^2), y) = \sigma\Big[z\big(2\Phi(z) - 1\big) + 2\varphi(z) - \tfrac{1}{\sqrt{\pi}}\Big], \quad z = (y - \mu)/\sigma, \tag{2.6}$$

giving the theoretical Gaussian-calibrated optimum $\text{CRPS}/\text{MAE} = 1/\sqrt{2} \approx 0.707$. The Probability Integral Transform [27] is uniform on $[0, 1]$ for a calibrated forecaster, with departures from uniformity flagged by the Kolmogorov–Smirnov statistic. The Winkler interval score [28] rewards narrow intervals and penalises non-coverage, enabling direct comparison of Gaussian, conformal, and adaptive conformal intervals. For binary error-detection, AUROC measures whether σ separates large-error from small-error nodes; AUPRC is more informative under class imbalance, with the random baseline equal to the positive-class prevalence [29, 30].

2.4. Surrogates and UQ in Adjacent Domains

GNN-based transport surrogates exploit the network topology that controls flow propagation [4]; PointNetTransfGAT (PointNet + transformer + graph attention) [4] is the base architecture used here. Existing UQ work on traffic targets temporal forecasting rather than the cross-sectional policy-impact setting studied here, and UQ for GNNs has been studied primarily for classification: Fuchsgruber et al. [31] propose single-pass epistemic uncertainty via Graph Energy-Based Models, and [32] explore uncertainty-guided active learning on graphs. Within conformal prediction, Romano et al. [20] introduce CQR (basis for Trials 10/11), Barber et al. [21] establish the impossibility of distribution-free conditional coverage, and the regression-calibration literature [24, 25, 33] provides the σ-scaling step used here.

3. Methodology

3.1. Provenance and Reused Components

This thesis builds on infrastructure provided by Natterer et al. [4] and adds the uncertainty-quantification layer on top. To keep the contribution transparent:

Reused (provided by Natterer et al.). The MATSim simulation pipeline that generated the 10,000 Paris policy scenarios; the PointNetTransfGAT architecture template (PointNetConv + TransformerConv + GATConv layers on the directed line graph); the five-feature node representation (`VOL_BASE_CASE`, `CAPACITY_BASE_CASE`, `CAPACITY_REDUCTION`, `FREESPEED`, `LENGTH`); and the dataset preprocessing convention.

Implemented and evaluated in this thesis. All 11 trials T1–T11 (trained from scratch on the 1,000-scenario subset) and the five-seed Deep Ensemble; the full UQ evaluation framework (MC Dropout at $S = 30$, S-convergence study, calibration audit via k_p, regression σ-scaling, split and adaptive conformal prediction, selective prediction, AUROC for top-error detection, CRPS / PIT / Winkler scoring); the three uncertainty-aware training extensions T9 (heteroscedastic head) and T10/T11 (CQR heads) including the frozen-vs-unfrozen-backbone comparison; the three non-GNN baselines (Random Forest, XGBoost, MLP) evaluated on the same scenario-level held-out test set; all experimental analyses, figures, tables and the resulting discussion.

3.2. Problem Formulation

The task is node-level regression on a directed line graph. Given a road network $\mathcal{G} = (V, E)$ with nodes $v \in V$ representing road segments and edges encoding adjacency, the goal is to predict the change in traffic volume at each segment after a policy intervention,

$$\Delta v_i = v_i^{\text{policy}} - v_i^{\text{baseline}}. \tag{3.1}$$

Targets are positive, negative, or zero; most road segments are unaffected by a typical capacity reduction. Each node carries five scalar features, `VOL_BASE_CASE`, `CAPACITY_BASE_CASE`, `CAPACITY_REDUCTION` (negative values denoting reductions), FREESPEED, LENGTH, together with the 2D start and end coordinates of the road segment, which are consumed by the PointNetConv layers below. Although both CAPACITY_REDUCTION (Table 3.1) and Δv contain a similar zero mass (88.7%), this should not be read as implying that only directly modified segments can change. Unmodified links can still experience volume changes through traffic rerouting and spillover effects in MATSim equilibrium [4]; the observed alignment of the two zero shares is an empirical property of this dataset and preprocessing, not a structural guarantee, and should be interpreted cautiously when reasoning about which segments are predictable.

Table 3.1.: Raw-unit statistics for the five input features. Mean and std are from the T8 StandardScaler fitted on the training partition (used for normalisation throughout the thesis); the median is from the 100-graph test set (n = 3,163,500 nodes). Following Natterer et al. [4], the HIGHWAY road-type feature is excluded so the models remain comparable to their baseline; any predictive signal from HIGHWAY is therefore dropped.

Feature	Mean	Std	Median	Notes
VOL_BASE_CASE	50.9	135.8	10.9	vehicles per hour, right-skewed
CAPACITY_BASE_CASE	1029	1264	480	vehicles per hour
CAPACITY_REDUCTION	−92.4	332.2	0	vehicles per hour, 88.7% zeros
FREESPEED	8.15	4.01	8.33	m/s, 6 discrete urban bands
LENGTH	91.6	109.9	58.4	metres, range 4.2 to 2569

3.3. Model Architecture

The surrogate uses the PointNetTransfGAT architecture introduced by Natterer et al. [4], which combines PointNet [11], transformer attention [34], and graph attention layers [10] into a single end-to-end model. Trial 1 uses a Linear(64→1) final layer; Trials 2–8 use GATConv(64→1), adding a final round of attention-weighted neighbour aggregation. Because Trial 1 was trained without dropout, MC Dropout is undefined for it (every stochastic forward pass produces an identical output).

The first two layers are PointNetConv layers and consume per-node start and end coordinates. The input data tensor also stores a midpoint coordinate, which is left in place to keep the preprocessing pipeline of Natterer et al. [4] identical, but the architecture used here does not consume it. For each pair of neighbours (i, j) the local MLP receives the relative displacement $\mathbf{pos}_j - \mathbf{pos}_i$ concatenated with the node feature vector; the first layer uses start-point geometry, and the second uses end-point geometry. This pair of layers encodes both the graph topology and the 2D spatial proximity of the segments. Two TransformerConv layers with four attention heads each follow, then a GATConv produces the 64-dimensional embedding fed into the final output layer (Table 3.2).

Table 3.2.: PointNetTransfGAT architecture. The input consists of node features $\mathbf{x} \in \mathbb{R}^{N\times 5}$ and positional coordinates $\mathbf{pos} \in \mathbb{R}^{N\times 3\times 2}$. The three coordinate slots in **pos** correspond to the segment start, end, and midpoint, but the layers below only consume the start and end coordinates.

Stage	Layer	Operation	In	Out
1	PointNetConv-1	Local MLP 7 $\to$ 256; global 256 $\to$ 512 $\to$ 512	5	512
2	PointNetConv-2	Local MLP 514 $\to$ 256; global 256 $\to$ 512 $\to$ 128	512	128
3	TransformerConv-1	4 heads, 64 dims/head	128	256
4	TransformerConv-2	4 heads, 128 dims/head	256	512
5	GATConv (mid)	1 head	512	64
6	GATConv / Linear	1 head (T2–T8) or pointwise (T1)	64	1

Dropout is applied within both PointNetConv MLPs and between the TransformerConv layers, gated by the `use_dropout` flag. For the uncertainty-aware extensions in Chapter 4, the 1,416,768-parameter T8 backbone is frozen (`requires_grad=False`) and only a small new output head (134 parameters in both T9 and T11) is optimised, motivated by the NLL–MSE trade-off (Section 2.3.3). Two subtleties of "frozen" deserve mention: T9 freezes the backbone *weights* but keeps backbone dropout active during head training and inference, so the backbone supplies epistemic uncertainty via stochastic forward passes through the head; T11 freezes the weights *and* forces backbone dropout off (single deterministic forward pass), since CQR does not require MC samples.

3.4. Uncertainty Quantification Methods

Six UQ *families* are compared here – MC Dropout, seed-ensemble variance, multi-model ensembles, Deep Ensembles, heteroscedastic regression (Trial 9), and CQR (Trials 10–11) – together with three post-hoc calibration and deployment tools that compose on top of

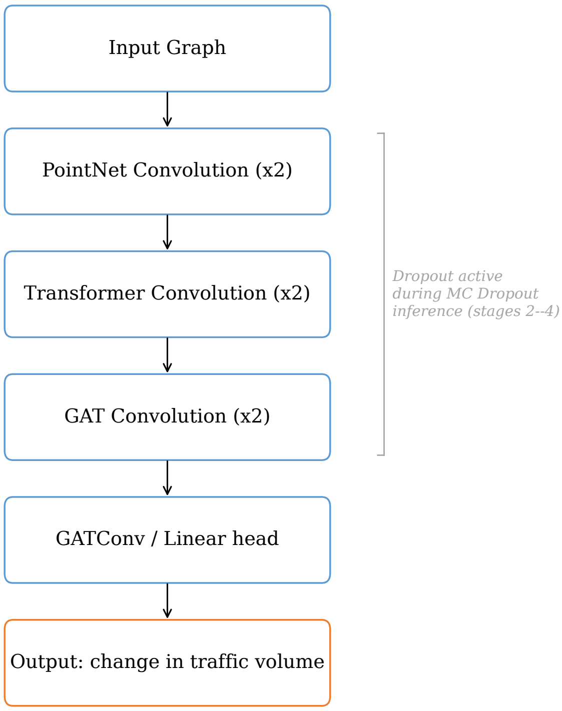

Figure 3.1.: Information flow through the PointNetTransfGAT architecture, top to bottom. The input graph is first encoded by two PointNetConv layers that extract local geometric features from each segment's start and end points. Two TransformerConv layers then capture longer-range relationships through self-attention across the network, two GATConv layers produce the final node embedding, and a final output layer (`GATConv(64→1)` for T2–T8 and `Linear(64→1)` for T1; see Table 3.2) produces the predicted change in traffic volume per node. Dropout, when enabled by the `use_dropout` flag (T2, T5, T6, T7, T8), is applied within the PointNetConv MLPs and between the TransformerConv layers and remains active during MC Dropout inference for uncertainty estimation.

the families: split and adaptive conformal prediction, regression σ-scaling, and selective prediction. The mathematical formulations and references appear in Sections 2.3.1–2.3.5 of Chapter 2; this section specifies the procedural choices made here.

MC Dropout. The trained model is evaluated $S = 30$ times with dropout active; per-node sample mean and standard deviation across passes form the prediction $\hat{\mu}$ and uncertainty $\hat{\sigma}$. A formal S-convergence study (Section 5.2) confirms that ρ has converged by $S = 30$, with the relative gain from $S = 30$ to $S = 50$ at $+1.03\%$.

Deep ensembles, single-architecture ensembles, and combined uncertainty. Five PointNetTransfGAT models are trained from scratch with seeds $\{42, 137, 256, 389, 512\}$ (Trial D); inference is deterministic and the inter-member spread is the uncertainty estimate. Two single-architecture variants are also evaluated: Experiment A averages MC Dropout over five inference seeds of the same T8 model; Experiment B is an R^2-weighted multi-model ensemble of T2/T5/T6/T7/T8. When MC Dropout and ensemble variance are both available, the two can be combined in quadrature, $\sigma_{\text{combined}} = \sqrt{\sigma_{\text{MC}}^2 + \sigma_{\text{ens}}^2}$.

Split conformal prediction (with an adaptive variant). The 100-graph T8 test set is partitioned into 50 calibration and 50 evaluation graphs (1,581,750 nodes each); the nonconformity score is $|y - \hat{y}|$ (or $|y - \hat{y}|/\sigma$ for the adaptive variant), the quantile uses the standard $\lceil (n+1)(1-\alpha) \rceil / n$ correction (Section 2.3.2), and intervals are constructed at $\{50, 70, 80, 90, 95\}\%$ coverage.

Post-hoc regression σ-scaling. A single scalar $T > 0$ rescales the predictive standard deviation, $\sigma_{\text{scaled}} = T \cdot \sigma_{\text{raw}}$, with the point prediction $\hat{\mu}$ left unchanged. This is the regression analogue of, not an instance of, classification temperature scaling [25]: that procedure rescales softmax logits ($z \mapsto z/T$) to correct over-confident probabilities, whereas the regression variant rescales the predictive standard deviation ($\sigma \mapsto T\sigma$) to correct under-dispersed intervals – same one-parameter, held-out, post-hoc structure operating on a different output channel. The rescaling is empirically warranted on the present continuous-valued Δv task because the raw MC Dropout posterior is too concentrated [16]: at $T = 1$ the nominal 1σ Gaussian interval covers only 32.7% of nodes against the 68.27% Gaussian target (Section 5.4), which is exactly the failure mode a single scalar $T > 1$ is designed to correct. Within this regression analogue, Kuleshov, Fenner and Ermon [24] establish the regression-calibration framework (extending Platt-style classification calibration [25] to regression by recalibrating the predictive distribution on a held-out set), Laves et al. [33, 35] specialise it to MC Dropout regression as a single-scalar σ-scaling, and Levi et al. [36] apply the same procedure to general regression heads. Under a Gaussian assumption on the recalibrated residuals, the maximum-likelihood estimator has the closed form $T^{\star 2}_{\mathrm{NLL}} = N^{-1} \sum_i (y_i - \hat{\mu}_i)^2 / \hat{\sigma}_i^2$; on the present data this Gaussian-NLL estimator returns $T^{\star}_{\mathrm{NLL}} \approx 6.35$ (Section 5.4), reflecting the heavy-tailed component of the absolute-residual distribution rather than a calibrated bulk fit. We therefore fit T by minimising the Kuleshov ECE [24] on a 30% random node-level subset of the test set (seed 42, 949,050 nodes) and evaluate on the remaining 70%. The ECE objective targets bulk-of-distribution coverage at 1σ rather than tail-weighted likelihood and is more robust to the heavy-tailed residuals observed here; the optimal value is $T^{\star} = 2.887$, which reduces ECE from 0.356 to 0.034 and brings 1σ coverage from 32.7% to 68.0% (Gaussian target 68.27%, Section 5.4).

Selective prediction. Predictions are sorted by σ ascending and the most confident fraction τ is retained; MAE is reported on the retained subset. This characterises how much accuracy can be sharpened when an abstention budget is available.

3.5. Evaluation Metrics

The evaluation follows the metrics defined in Chapter 2. Point-prediction quality is measured by R^2, MAE, and RMSE. Uncertainty ranking quality is measured by Spearman ρ between σ and $|y - \hat{y}|$. Calibration is reported via the Kuleshov ECE [24], the conformal coverage probability $\mathrm{PICP}_{1-\alpha}$, and the empirical scaling factor

$$k_p = \mathrm{Quantile}\big(\{|y_i - \hat{y}_i| / (\sigma_i + \varepsilon)\}_{i=1}^{n}\,;\; p\big)\,, \tag{3.2}$$

where $\varepsilon = 10^{-8}$ is a small numerical stability constant that avoids division by zero when σ_i is close to zero. For a perfectly calibrated Gaussian, $k_{0.95} = 1.96$. Proper scoring rules, CRPS (Equation (2.6)), the PIT, and the Winkler interval score, including the PIT KS statistic, are reported on the full 3,163,500-node test set. Error detection is reported via AUROC at the top-10% and top-20% error thresholds, with AUPRC as the imbalance-aware companion.

4. Experimental Setup

4.1. Dataset and Splits

The dataset is a MATSim simulation of the Paris metropolitan area provided by Natterer et al. [4]. Each scenario corresponds to a distinct policy intervention: a random subset of road segments is selected and assigned a capacity reduction in $\{10\%, 20\%, \ldots, 100\%\}$. MATSim is then run from the same baseline plans, and the per-segment difference in hourly volume is recorded as the supervised target Δv (Table 4.1).

Table 4.1.: Paris MATSim dataset. All experiments use the 1,000-scenario subset.

Property	Value
Road segments per graph	31,635
Available scenarios (full)	10,000
Scenarios used (10% subset)	1,000
Test scenarios (T1–T6 split 80/15/5)	50
Test scenarios (T7–T11, DE; split 80/10/10)	100
Test predictions, T7–T11, DE	3,163,500

Trials T1–T6 use an 80/15/5 split (800/150/50 scenarios); Trials T7–T11 and the Deep Ensemble use 80/10/10 (800/100/100), with the larger 100-graph test set enabling the high-resolution UQ analyses in Chapter 5. The split is at the scenario level (each scenario appears in exactly one split) with split seed 42, ensuring identical scenario assignment across all trials that share a split configuration.

A note on the 10% subset. The full Natterer corpus is around 200 batch files of 50 graphs each ($\approx$ 10,000 scenarios); we load the first 20 batches, which gives the 1,000-scenario subset reused across all eleven trials so that method-to-method differences are evaluated on identical data. The 100-graph evaluation set (3,163,500 node-level predictions per trial) is large enough for stable per-graph statistics: per-graph Spearman ρ has std 0.023, with a bootstrap 95% CI of ± 0.005 on the mean. Possible subset-selection bias from the contiguous batch ordering, and the question of whether the findings survive at the full 10,000-scenario scale, are discussed under threats to validity in Section 6.10.

4.2. Training Protocol

Eight base trials of the PointNetTransfGAT architecture are evaluated. Trial 1 uses a `Linear` final layer; T2–T8 use `GATConv`. Hyperparameters are listed in Table 4.2.

Table 4.2.: Training trials. T1 uses a Linear final layer; T2–T8 use GATConv.

Trial	Batch	Split	Effective dropout	LR	Weighted MSE	Final layer
T1	32	80/15/5	0.0†	10^{-3}	No	Linear
T2	16	80/15/5	0.3	5×10^{-4}	No	GATConv
T3	16	80/15/5	0.0	5×10^{-4}	Yes	GATConv
T4	16	80/15/5	0.0	5×10^{-4}	Yes	GATConv
T5	8	80/15/5	0.3	5×10^{-4}	No	GATConv
T6	8	80/15/5	0.3	3×10^{-4}	No	GATConv
T7	8	80/10/10	0.3	6×10^{-4}	No	GATConv
T8	**8**	**80/10/10**	**0.2**	$\mathbf{5 \times 10^{-4}}$	**No**	**GATConv**

† T1 was instantiated with `use_dropout=False`, so the effective dropout rate is 0; MC Dropout is therefore undefined on T1. T3 and T4 likewise have effective dropout zero and are not MC-Dropout-compatible.

All trials use the AdamW optimiser [37] with weight decay 10^{-4} and MSE loss. T3 and T4 use a weighted variant $\mathcal{L}_{\text{weighted}} = \frac{1}{n} \sum_i w_i (y_i - \hat{y}_i)^2$, with weights w_i proportional to $|y_i|$ so that non-zero-Δv examples carry more influence. T3 and T4 share identical hyperparameters; T4 is a replication run of T3 with a different random seed, included to check that the weighted-MSE failure mode is systematic rather than a one-off. The implementation uses PyTorch [38] and PyTorch Geometric [39]; hardware is described in Section 4.5. The early-stopping criterion is validation R^2 with a patience of 25 epochs and a minimum improvement of 10^{-3}. Among Trials 2–8, T8 attains the highest test $R^2 = 0.5957$ on the 100-graph evaluation set; together with active dropout ($p = 0.2$), this makes it the strongest UQ-compatible base in this exploratory comparison, and it is used as the primary model for the UQ analyses that follow.

4.3. Post-Hoc UQ and Ensemble Experimental Design

Post-hoc analyses on Trial 8 and the ensemble experiments are summarised in Table 4.3. All operate on the same 100-graph T8 test set (3,163,500 nodes) unless noted.

Table 4.3.: Post-hoc UQ and ensemble experimental design. Scenario-level splits use seed 42 throughout.

Experiment	Protocol
MC Dropout (primary)	$S = 30$ stochastic forward passes; primary metric Spearman ρ between σ and $\lvert y - \hat{y} \rvert$.
S-convergence	10 test graphs, $S \in \{5, 10, \ldots, 50\}$.
Bootstrap CI	Per-graph ρ resampled $B = 10{,}000$ times for 95% CI on the mean.
T7 replication	Same MC Dropout / AUROC / conformal protocol on Trial 7.
Split conformal	Scenario-level 50/50 partition; nonconformity $\lvert y - \hat{y} \rvert$; coverage at $\{0.5, 0.7, 0.8, 0.9, 0.95\}$. Adaptive variant normalises by MC Dropout σ.
Calibration audit	Empirical k_p at $p \in \{0.5, 0.7, 0.8, 0.9, 0.95\}$ on the full test set.
σ-scaling	Fitted on a 30% random node-level subset (949,050 nodes), evaluated on the remaining 70%; grid $T \in [0.5, 5.0]$, ECE-minimising.
Scoring + selective + AUROC	CRPS, PIT, Winkler on the full test set; selective prediction at $\tau \in \{0.10, 0.25, 0.50, 0.75, 1.00\}$; AUROC/AUPRC at top-10% / top-20%.
Experiment A (seed ensemble)	Same trained T8, 5 inference seeds; compares σ_{MC}, $\sigma_{\text{seed-ens}}$, and the quadrature combination.
Experiment B (multi-model)	R^2-weighted ensemble of T2/T5/T6/T7/T8, deterministic inference.
Trial D (Deep Ensemble)	5 PointNetTransfGAT trained from scratch with seeds $\{42, 137, 256, 389, 512\}$, T8 hyperparameters, 80/10/10 split.

4.4. Uncertainty-Aware Training Extensions

Three trials evaluate whether uncertainty-aware training objectives can produce natively calibrated intervals while preserving T8's accuracy. All replace T8's final `GATConv(64→1)` with a 134-parameter `GATConv(64→2)` head, differing in head interpretation, loss, and backbone trainability (Table 4.4). Other hyperparameters (optimiser AdamW, learning rate 5×10^{-4}, batch 8, 80/10/10 split) inherit from T8. T9 freezes the backbone weights but keeps dropout active for MC Dropout epistemic estimation ($\sigma_{\text{tot}} = \sqrt{\sigma^2_{\text{alea}} + \sigma^2_{\text{epi}}}$); T10/T11 use a single deterministic forward pass.

Table 4.4.: Uncertainty-aware training extensions. Head output channels are $(\hat{\mu}, \log \hat{\sigma}^2)$ for T9 and $(\hat{q}^{\text{lo}}, \hat{q}^{\text{hi}})$ for T10/T11 ($\tau = 0.05, 0.95$).

Trial	Loss	Backbone	UQ output	Gates
T9	Hetero NLL ($\lambda =$ 0.1) [22]	Frozen	MC + $\hat{\sigma}_{\text{alea}}$	$R^2 \geq 0.55$, $\text{PICP}_{90} \geq 85\%$, $\text{PICP}_{95} \geq 90\%$
T10	Pinball	Unfrozen	CQR + $\hat{Q}$	$R^2 \geq 0.57$, $\text{PICP}_{90} \geq 88\%$, $\text{PICP}_{95} \geq 93\%$, $\hat{Q} > 0$, $W_{90} < W_{95}$
T11	Pinball	Frozen	CQR + $\hat{Q}$	same as T10

The single design difference between T10 and T11 is the backbone-trainability flag; head, loss, calibration protocol and learning rate are identical, so the T10/T11 contrast isolates that flag.

4.5. Computational Resources

All training and inference used a single NVIDIA T4 GPU (16 GB VRAM). Wall-clock runtimes from the saved `training_log.json` files: the eight base trials totalled ≈ 13 h; T8 MC Dropout inference ($S = 30$, 100 graphs) 228 min; Trials 9, 10 and 11 took 14.5, 87 and 39.7 h respectively; the Deep Ensemble $\approx 5\times$ a single T8 run. Trial 10 is the longest because it fine-tunes the full 1,416,768-parameter backbone end-to-end, while Trials 9 and 11 train only a 134-parameter head with the backbone frozen. Non-GNN baselines (Section 5.11) were run separately on an A100 (80 GB) runtime.

5. Results

Every numeric value reported in this chapter has been recomputed from the saved per-trial prediction arrays or read from the canonical metric files for that trial. The chapter follows the natural order: base trial accuracy first, then MC Dropout uncertainty quality, calibration and conformal coverage, post-hoc and proper-scoring diagnostics, ensembles, the three uncertainty-aware extensions, and finally the stratified analysis.

Where the method is weakest – read this first. Before reporting the headline numbers it is worth flagging the regime in which the proposed UQ pipeline is least reliable, because that regime is also the most policy-relevant. Section 5.10 stratifies the test set by $|\Delta v|$ and shows that the ranking quality of MC Dropout weakens substantially on the largest-$|\Delta v|$ quartile (Spearman ρ falls from 0.721 on Q1 to 0.100 on Q4, while MAE rises from 1.24 to 10.08 veh/h). The high Q1 number is partly mechanical, but the Q4 degradation is real: on the segments where the policy intervention has the strongest effect, the uncertainty signal is the least informative. For decision support this is the opposite of the regime a planner cares about. Because $|\Delta v|$ itself is unobservable at deployment, the recommendations in Section 5.12 use the two observables that are available – predicted uncertainty σ and the magnitude of the predicted response $|\widehat{\Delta v}|$ – as proxies, and treat high-uncertainty predictions and predictions with large predicted responses as cases that should be re-simulated in MATSim rather than trusted directly.

5.1. Base Trial Performance (T1–T8)

Table 5.1 summarises test-set performance. T1–T6 are evaluated on 50 test graphs under the 80/15/5 split, while T7 and T8 use 100 graphs under the 80/10/10 split (3,163,500 nodes). Because the test sets differ, comparisons across the two groups carry a component that comes from test-set composition rather than from architectural choice, and should be read descriptively. Among the trials that combine the GATConv output with non-zero dropout (T2, T5, T6, T7 and T8), T8 is the strongest with $R^2 = 0.5957$, MAE = 3.957 veh/h and RMSE = 7.118 veh/h. T3 and T4 disable dropout entirely and are therefore not MC-Dropout-compatible. Trial 1 reaches the highest absolute $R^2 = 0.7860$ but its effective dropout is zero, which makes MC Dropout (the main UQ method studied in the thesis) undefined on it. The two weighted-loss runs (T3 and T4) fail to generalise at this data scale, with R^2 between 0.22 and 0.24. The T1-versus-T8 gap is discussed in Section 6.9.

Table 5.1.: Test-set performance, recomputed from the saved `.npz` files.

Trial	Final layer	Split	Test graphs	R^2	MAE	RMSE
T1	Linear	80/15/5	50	0.7860	2.972	5.396
T2	GATConv	80/15/5	50	0.5117	4.328	8.151
T3	GATConv	80/15/5	50	0.2246	5.990	10.270
T4	GATConv	80/15/5	50	0.2426	6.080	10.151
T5	GATConv	80/15/5	50	0.5553	4.242	7.778
T6	GATConv	80/15/5	50	0.5223	4.324	8.061
T7	GATConv	80/10/10	100	0.5471	4.060	7.534
T8	**GATConv**	**80/10/10**	**100**	**0.5957**	**3.957**	**7.118**

T1, T8 and the Deep Ensemble at a glance. Table 5.2 sets out the trade-off behind the choice of T8 as the main UQ model. T1 has the highest raw test R^2 in the project but is not compatible with MC Dropout. T8 is the strongest UQ-compatible base, which is the property that matters for the rest of the thesis, but it is not the highest-accuracy model; the Deep Ensemble is. Reading the rest of this chapter as a statement about UQ quality on T8, rather than as a statement about the best possible point-prediction surrogate, is therefore the right framing.

5.2. MC Dropout Uncertainty (T8)

MC Dropout with $S = 30$ on the full T8 test set produces pooled Spearman $\rho = 0.4820$ between σ and $|y - \hat{y}|$, with mean $\bar{\sigma} = 1.369$ veh/h. The pooled value includes the Q1 quartile of zero-effect segments, where the correlation is partly mechanical (Section 5.10), so the per-graph mean 0.464 with bootstrap 95% CI $[0.460, 0.469]$ is the more conservative summary. MC-mean accuracy is approximately 0.01 R^2 below the deterministic forward pass ($R^2 = 0.5856$ vs 0.5957, a difference of 0.0101; MC-mean MAE $=$ 3.948 vs deterministic 3.957 veh/h), a small expected effect of stochastic

Table 5.2.: Compact comparison of T1, T8 and the Deep Ensemble. T1 has the highest absolute R^2 but is not MC-Dropout-compatible; T8 is the primary UQ base; the Deep Ensemble has the highest R^2 among the UQ-compatible models. T1 was evaluated on a separate 50-graph test set under an 80/15/5 split, whereas T8 and the Deep Ensemble share a different 100-graph test set under the 80/10/10 split (not a subsampled version of the same set), so the two test sets are independent and the cross-group comparison should be read descriptively.

Model	Test graphs	R^2	MC-Dropout-compatible?	Role in this thesis
T1	50	**0.7860**	No (effective dropout 0)	Architectural reference; not used for UQ
T8	100	0.5957	Yes	Primary UQ base model
Deep Ensemble	100	**0.6841**	Yes (per member)	Strongest UQ-capable GNN point accuracy

averaging.[1]

Per-graph stability. Per-graph ρ has mean 0.464, std 0.023, range $[0.407, 0.505]$. A bootstrap ($B = 10{,}000$) gives a 95% CI of $[0.460, 0.469]$ for the mean per-graph ρ.

***S*-convergence.** Convergence on 10 test graphs: $S = 30 \to 50$ adds only $+1.03\%$ in ρ ($0.466 \to 0.471$); $S = 5 \to 30$ contributes $+10.8\%$. $S = 30$ is therefore used throughout.

5.3. Calibration and Conformal Prediction (T8)

A note on what coverage means in this section. Coverage statements below are *marginal*: PICP is the proportion of test residuals that fall inside the predicted interval averaged across scenarios and nodes on the held-out evaluation split. It is not a conditional guarantee for any single scenario or any single uncertainty bin, and node-level predictions inside one graph remain spatially correlated. The conditional-coverage range across uncertainty deciles is reported separately later in this section, and the exchangeability assumption that backs the marginal guarantee is examined in Section 6.4.

The empirical scaling factor k_p (Equation (3.2)) quantifies the mismatch between raw MC Dropout σ and a calibrated Gaussian (Table 5.3). At the 95% nominal level, $\hat{y} \pm 1.96\sigma$ covers only 54.85% of residuals: raw MC Dropout is useful as a ranking signal but not as an interval estimator.

[1] Cross-trial replication on Trial 7 (dropout 0.3): $\rho = 0.4437$, $k_{95} = 16.15$, AUROC $= 0.7416$ at top-10% – qualitatively unchanged.

Table 5.3.: T8 empirical k_p vs Gaussian reference, full 3,163,500-node test set.

Nominal p	Empirical k_p	Gaussian k_p	Cov. at Gaussian k_p
50%	1.71	0.67	23.4%
70%	3.09	1.04	33.7%
80%	4.55	1.28	40.1%
90%	7.74	1.64	48.6%
95%	**11.66**	**1.96**	**54.85%**

Split conformal prediction with 50 calibration and 50 evaluation graphs gives $q_{90} = 9.920$ with empirical $\text{PICP}_{90} = 90.02\%$ and $q_{95} = 14.677$ with $\text{PICP}_{95} = 95.01\%$. The empirical marginal coverage on the held-out evaluation graphs is therefore close to the nominal targets. Exchangeability, on which the formal conformal coverage guarantee rests, is enforced here by the scenario-level calibration/evaluation split; node-level dependence within a single graph is not removed by that split and is what limits per-scenario (conditional) coverage, discussed in Section 6.4. Adaptive conformal uses MC Dropout σ as a normaliser and yields $q^{\text{adapt}} = 7.71$ at the 90% level, with empirical global $\text{PICP}_{90} = 89.87\%$, 0.13 percentage points below the nominal target.

Conditional coverage. Standard conformal can over-cover the low-σ nodes and under-cover the high-σ nodes; here this shows up as 98.1% coverage on decile D1 and only 59.0% on D10, which is consistent with the impossibility result of Barber et al. [21] for distribution-free conditional coverage. Adaptive conformal narrows the observed range across deciles to $[83.7\%, 96.4\%]$.

5.4. Post-Hoc Regression σ-Scaling

Motivation. The calibration audit identifies the bulk of the underdispersion as a scale shift in σ, the failure mode a single-scalar rescaling is designed to correct [24, 33, 36]; Foong et al. [16] prove the MC Dropout posterior is too concentrated relative to the exact Bayesian posterior. The residual ratios $|y - \hat{\mu}|/\sigma$ are heavy-tailed (median 1.71, $p_{99} = 23.6$, max > 900), so the Gaussian-NLL closed form $T^{\star 2}_{\text{NLL}} = N^{-1} \sum (y - \hat{\mu})^2/\sigma^2$ returns $T \approx 6.35$ and over-compensates (1σ coverage at 86.8%). The ECE objective targets bulk coverage and is tail-robust; $T^{\star}_{\text{ECE}} = 2.887$ hits the 68.3% Gaussian target within 0.3 pp, and the residual tail mismatch is handled by conformal prediction (Section 5.5).

Fitting a scalar T on a 30% random node-level calibration subset (seed 42, 949,050

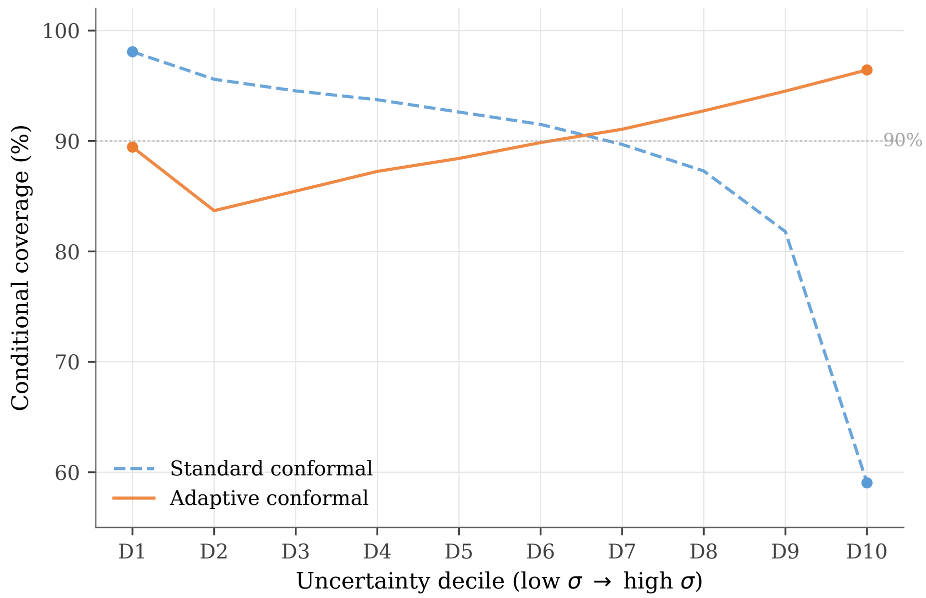

Figure 5.1.: Standard conformal drifts 39 pp across deciles; adaptive narrows to ~ 13 pp.

nodes) gives $T^\star = 2.887$. Because this split is at the node level, the calibration and evaluation halves are not strictly independent at the scenario level: nodes within the same graph remain correlated. The diagnostics in Table 5.4 should therefore be read as empirical post-fit coverage rather than as a guarantee at the graph level. ECE drops from 0.356 to 0.034, a 90.5% reduction; the 1σ coverage rises from 32.7% to 68.0%, which is within 0.3 pp of the 68.3% Gaussian target; and k_{95} falls from 11.66 to 4.04. Both before- and after-scaling values of k_{95} are computed on the same held-out 70% evaluation subset, so the comparison is consistent. The remaining gaps at 2σ and 3σ reflect heavier-than-Gaussian tails: a single scalar can rescale the predictive distribution but cannot reshape it.

5.5. Proper Scoring Rules, Selective Prediction, Error Detection

The proper-scoring-rule diagnostics on the full T8 test set are as follows. Mean CRPS is 3.384 veh/h, giving CRPS/MAE $= 0.857$ against the Gaussian-calibrated optimum $1/\sqrt{2} \approx 0.707$ (the MAE in the denominator is the MC-mean MAE 3.948 veh/h from Section 5.2, since CRPS is computed against the MC predictive distribution). The PIT has mean 0.433 (ideal 0.500) and KS statistic 0.245 (ideal 0). Winkler interval

Table 5.4.: Regression σ-scaling. Gaussian targets: 68.27%, 95.45%, 99.73%.

Metric	Before ($T = 1$)	After ($T = 2.887$)	Δ
ECE	0.356	**0.034**	−90.5%
Coverage at 1σ	32.7%	**68.0%**	+35.3 pp
Coverage at 2σ	55.6%	85.0%	+29.4 pp
Coverage at 3σ	69.1%	91.6%	+22.5 pp
k_{95}	11.66	**4.04**	−7.62

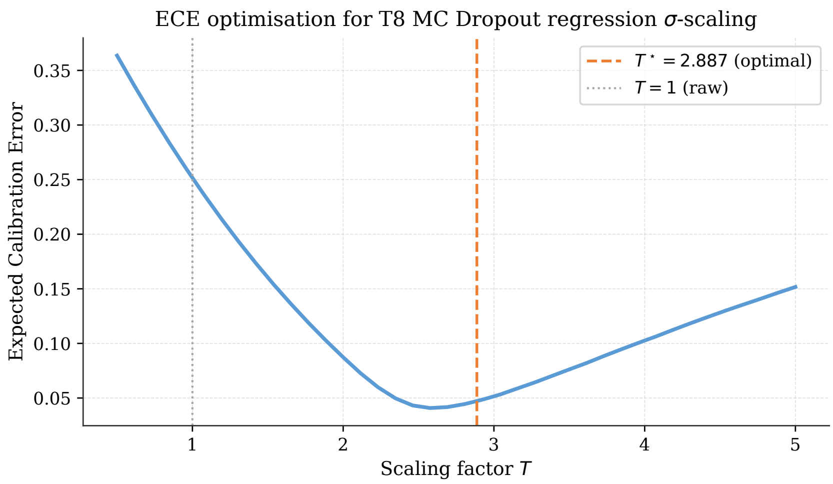

Figure 5.2.: ECE as a function of the scaling factor T on the calibration subset; the optimum at $T^\star = 2.887$ is marked. Coverage and reliability numbers are in Table 5.4.

scores at the 90% level are 49.68 for raw MC, 35.78 for standard conformal (−28.0%), and **32.32** for adaptive conformal (−35.0% vs raw). The PIT KS drops to 0.104 after regression σ-scaling, but the residual deviation reflects an upward bias in $\hat{\mu}$ that a scalar rescaling cannot fix: regression σ-scaling expands the predictive variance symmetrically around $\hat{\mu}$ and does not shift the predictive centre. Conformal prediction is the natural complementary tool here because its intervals expand around $\hat{\mu}$ to contain the observed residuals, regardless of where $\hat{\mu}$ itself sits.

Selective prediction. Filtering by MC Dropout σ percentile and retaining only the most confident fraction yields the curve in Figure 5.3. The 50% retention point reduces MAE by 41.2% to 2.32 veh/h; at 25% retention the reduction is 54.5%, and at 10% it is 73.4%.

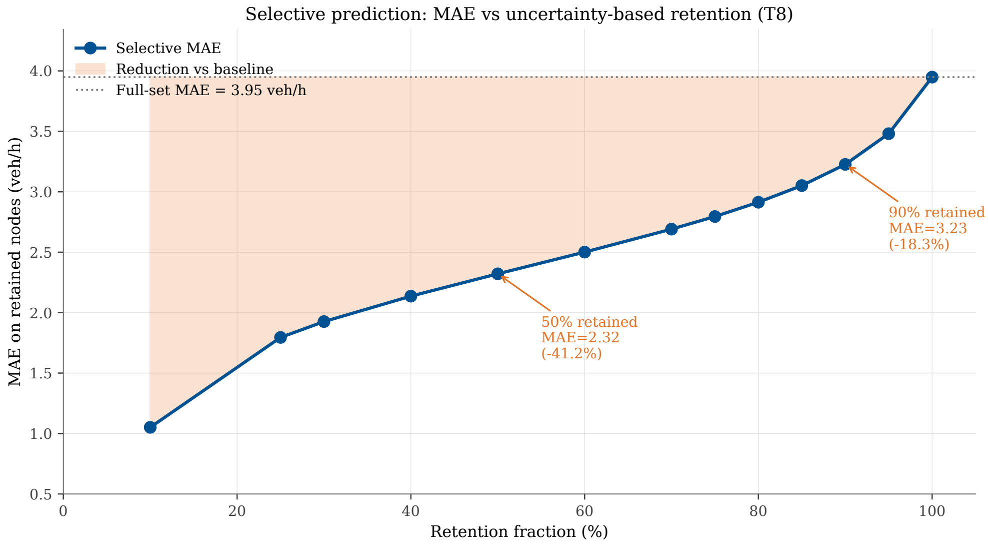

Figure 5.3.: Selective prediction curve. Annotations mark 50% and 90% retention.

Error detection. Treating σ as a classifier of large errors gives an AUROC of 0.7548 at the top-10% threshold and 0.7324 at the top-20%, both computed on the full 3,163,500-node test set. Both metrics replicate on Trial 7 (0.7416 and 0.7151; see the footnote in Section 5.2). One caveat for both AUROC and the selective-prediction MAE reduction is the high zero-mass in the target: 88.7% of test nodes have $\Delta v = 0$, so a model that predicts near zero with low σ is trivially correct for the bulk of the test set; the effect is that AUROC and MAE-on-retained-nodes both look stronger than they would on a balanced subset, and the policy-relevant Q4 stratum (Section 5.10) remains the more demanding regime.

5.6. Ensemble Experiments

Single-architecture ensembles (Experiments A, B). MC Dropout averaged over 5 inference seeds (Experiment A) gives $\rho = 0.4908$; the seed-ensemble variance gives $\rho = 0.4370$ ($\approx$ 11.0% relative drop from MC averaging); their quadrature combination gives $\rho = 0.4909$, indistinguishable from MC Dropout alone. The multi-model ensemble of T2/T5/T6/T7/T8 (Experiment B, R^2-weighted) gives $\rho = 0.4333$ and $R^2 = 0.5656$, below the best single model (T8, $R^2 = 0.5957$) because averaging across models of uneven quality dilutes the strongest predictor.

Deep Ensemble (Trial D). Five PointNetTransfGAT models with T8 hyperparameters and seeds $\{42, 137, 256, 389, 512\}$ form the strongest GNN-based UQ-capable point predictor: $R^2 = \mathbf{0.6841}$ (+14.8% over T8), MAE = 3.485, RMSE = 6.293 veh/h. Member R^2 values lie in $[0.640, 0.650]$, so the ensemble mean exceeds every member. The uncertainty signal is weaker than MC Dropout: $\rho = 0.3997$ (-17.1%), $k_{95} = 15.18$. Chapter 6 discusses this trade-off.

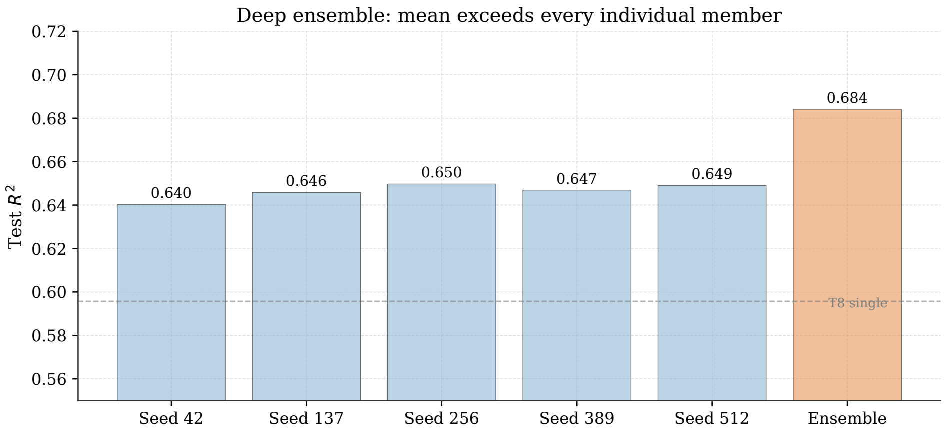

Figure 5.4.: Deep Ensemble: mean exceeds every individual member.

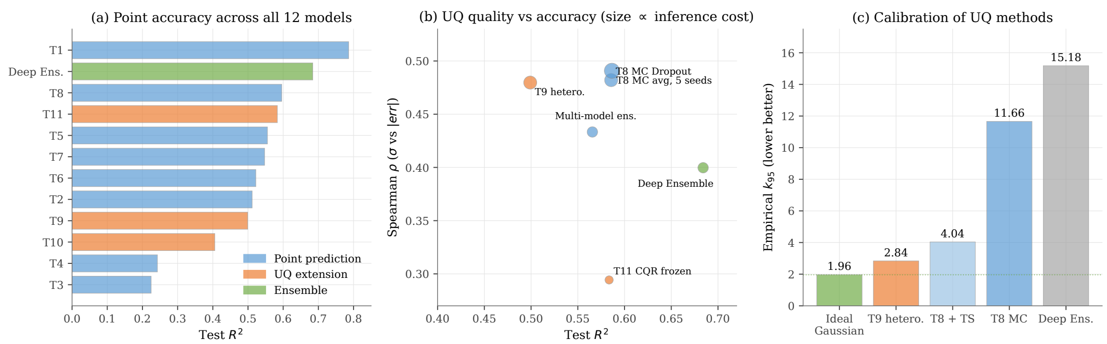

Figure 5.5.: Master summary of the twelve GNN-based and uncertainty-aware models evaluated here. Panel (a) sorts trials by test R^2 (point-prediction trials blue, uncertainty-aware extensions orange, Deep Ensemble green); the Deep Ensemble leads UQ-compatible models at $R^2 = 0.6841$. Panel (b) is the accuracy/UQ trade-off (R^2 vs Spearman ρ, marker size proportional to log inference cost): Deep Ensemble and MC Dropout sit on opposite corners; T11 (frozen-backbone CQR) occupies an attractive low-cost corner. Panel (c) compares k_{95} (lower is better, Gaussian ideal 1.96); T9 is closest to ideal among trained models.

5.7. Trial 9: Heteroscedastic, Frozen Backbone

T9 freezes the T8 backbone and trains only the 134-parameter heteroscedastic head under Gaussian NLL. Two of three gates pass: $\text{PICP}_{90} = 86.90\%$, $\text{PICP}_{95} = 90.01\%$, but $R^2 = 0.4991$ misses the ≥ 0.55 gate (deterministic single pass: $R^2 = 0.5118$). k_{95} improves $4\times$ to 2.84. The R^2 miss is consistent with the NLL–MSE trade-off (Section 2.3.3): with a small head, the gradient is reduced more cheaply by inflating $\hat{\sigma}$ than by improving $\hat{\mu}$ [22]. The decomposition attributes 99.85% of nodes to the aleatoric component ($\bar{\sigma}_{\text{alea}}/\bar{\sigma}_{\text{epi}} = 4.24$). The configuration here, a 134-parameter head trained by Gaussian NLL on top of a frozen 1,416,768-parameter backbone, is exactly the small-head regime that Seitzer, Tavakoli, Antic, and Martius [22] identify as prone to $\hat{\sigma}$-inflation; the NLL-driven explanation is therefore a priori more likely than genuine aleatoric dominance, and separating the two would require ground-truth uncertainty labels that are not available here.

Table 5.5.: Trial 9 versus T8 and go/no-go gates. The k_{95} row is reported for diagnostic comparison with T8; it is not one of the three Trial 9 gates (R^2, PICP_{90}, PICP_{95}).

Metric	T8	T9	Gate	Status
R^2 (MC mean)	0.5957	**0.4991**	≥ 0.55	**FAIL**
PICP_{90}	90.02%	86.90%	$\geq 85\%$	PASS
PICP_{95}	95.01%	90.01%	$\geq 90\%$	PASS
k_{95} (diagnostic)	11.66	**2.84**	lower better	$4\times$ improvement

Table 5.6.: Trial 9 attributes nearly all predicted uncertainty to the aleatoric component. The small-head/NLL configuration here is the regime in which Seitzer, Tavakoli, Antic, and Martius [22] specifically predict $\hat{\sigma}$-inflation, so the NLL-driven explanation is a priori more likely than genuine aleatoric dominance; separating the two would require ground-truth uncertainty labels not available here.

Quantity	Value
Mean aleatoric $\bar{\sigma}_{\text{alea}}$ (veh/h)	4.66
Mean epistemic $\bar{\sigma}_{\text{epi}}$ (veh/h)	1.10
Ratio $\bar{\sigma}_{\text{alea}}/\bar{\sigma}_{\text{epi}}$	4.24
Fraction of nodes that are aleatoric-dominated	99.85%

5.8. Trial 10: CQR with Unfrozen Backbone (Negative)

Trial 10 fine-tunes the full backbone under the pinball loss at $\tau \in \{0.05, 0.95\}$, inheriting T8's learning rate (5×10^{-4}) and the same head, loss, calibration protocol and 80/10/10 split as Trial 11. The midpoint $R^2 = 0.4057$ is reported in Table 5.7; three of the six gates fail. Of the three failing gates, only the R^2 miss (0.4057 vs ≥ 0.57) is decisive; PICP_{95} misses by 1.2 percentage points (91.78% vs 93%) and $\hat{Q}_{90} = -0.0011$ is at the rounding boundary of the > 0 threshold. The R^2 collapse is therefore the substantive failure mode of T10, and the pattern is consistent with pinball gradients reshaping representations that were originally learned for the conditional mean under MSE.

Table 5.7.: T10 against the six CQR gates.

Metric	T8	T10	Gate	Status
R^2 (midpoint)	0.5957	**0.4057**	≥ 0.57	**FAIL**
PICP_{90}	90.02%	89.47%	$\geq 88\%$	PASS
PICP_{95}	95.01%	**91.78%**	$\geq 93\%$	**FAIL**
Width_{90} < Width_{95}	—	$17.78 < 20.02$	yes	PASS
$\hat{Q}_{90}$	—	-0.0011	> 0	**FAIL**
$\hat{Q}_{95}$	—	1.121	> 0	PASS

5.9. Trial 11: CQR with Frozen Backbone (Positive)

T11 freezes the T8 backbone and trains only the 134-parameter quantile head with the same pinball loss, calibration protocol and learning rate as T10. All six gates pass (Table 5.8): $R^2 = 0.5835$ ($\approx 2\%$ below T8), both PICP targets hit to within 0.2 pp, both conformal corrections positive, widths monotone. MAE rises slightly to 4.302 vs T8's 3.957 – the cost of asymmetric calibrated intervals at one deterministic forward pass. All other settings are held fixed across T10 and T11, so the contrast isolates the backbone-trainability flag and gives the strongest evidence in the chapter for the freezing-backbone observation developed in Chapter 6.

5.10. Stratified UQ by $|\Delta v|$ Quartile

Rank-based quartiles of $|\Delta v|$ (790,875 nodes each) show a sharp regime dependence. Q1 contains the segments where the policy has no effect at all ($|\Delta v| = 0$ throughout), and Q4 contains the segments with the largest absolute response, up to about 230

Table 5.8.: T11 passes all six CQR gates.

Metric	T8	T11	Gate	Status
R^2 (midpoint)	0.5957	**0.5835**	≥ 0.57	**PASS**
PICP_{90}	90.02%	**89.82%**	$\geq 88\%$	**PASS**
PICP_{95}	95.01%	**94.91%**	$\geq 93\%$	**PASS**
$\text{Width}_{90} < \text{Width}_{95}$	—	$17.83 < 24.82$	yes	**PASS**
$\hat{Q}_{90}$	—	**0.124**	> 0	**PASS**
$\hat{Q}_{95}$	—	**3.623**	> 0	**PASS**

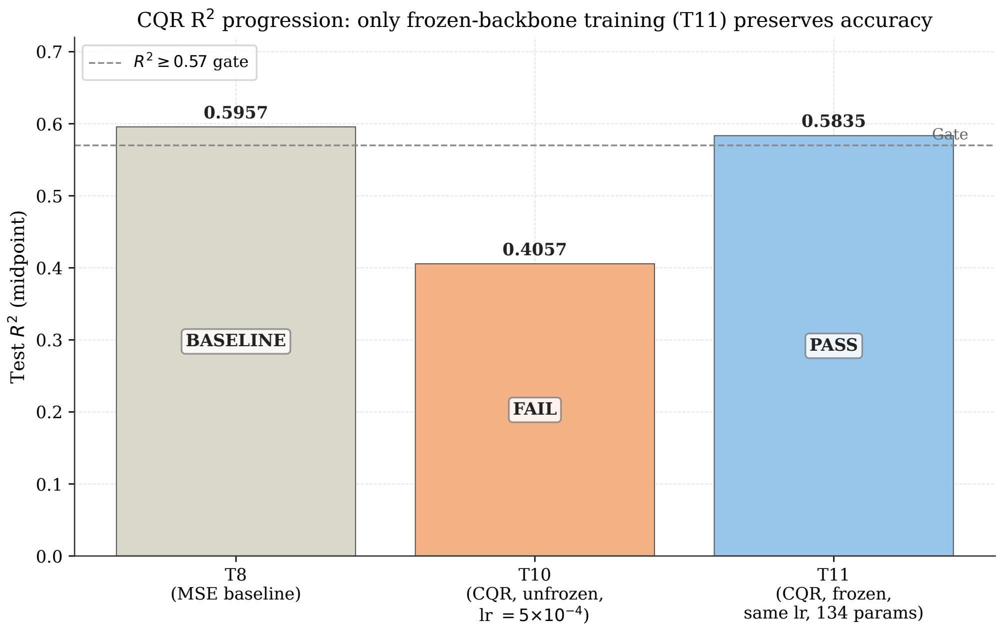

Figure 5.6.: Coefficient of determination across the CQR trials. Trial 8 (the MSE-trained baseline) achieves $R^2 = 0.596$. Trial 10 with the full backbone unfrozen drops to $R^2 = 0.4057$ and fails the $R^2 \geq 0.57$ gate; Trial 11 with the backbone frozen recovers to $R^2 = 0.5835$ and meets the gate. Because all other settings are held fixed across T10 and T11, the figure isolates backbone trainability as the only design difference and supports the freezing-backbone observation in this experimental setting.

veh/h. Within-quartile Spearman ρ between σ and $|\text{error}|$ falls from 0.721 on Q1 to 0.100 on Q4 (Figure 5.7), and MAE rises from 1.24 to 10.08 veh/h. The high Q1 ρ is partly a mechanical artefact: when $|\Delta v|$ is exactly zero, both $|\text{error}|$ and σ reduce to functions of the model's small output magnitude on unchanged segments, and their correlation reflects that shared dependence as much as a genuine ranking signal. The substantive degradation on Q4 is real, and the contributing mechanisms are discussed in Section 6.3.

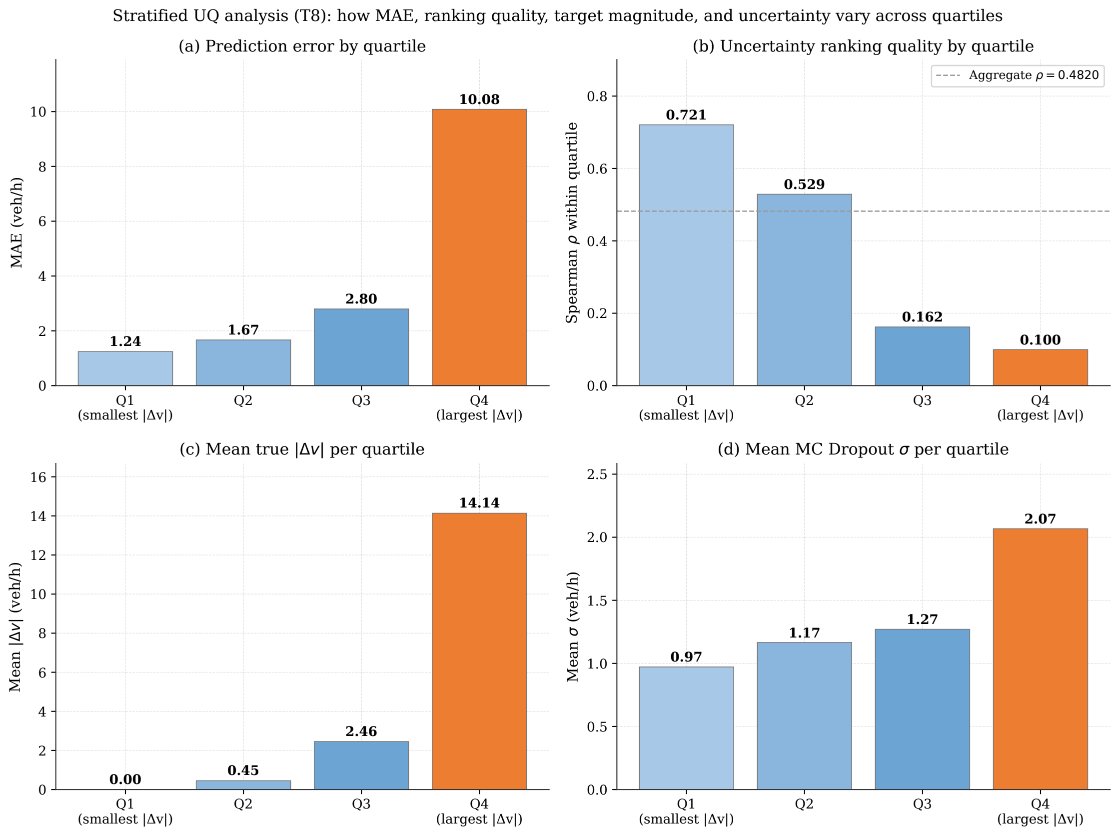

Figure 5.7.: Stratified UQ: MAE rises and ρ falls as $|\Delta v|$ grows.

5.11. Non-GNN Baselines

Three non-GNN baselines on the same five node features and the same scenario-level held-out test set as T8: Random Forest (cuML [40] defaults, `n_estimators` = 100, max depth 16 [41]), XGBoost (GPU, default hyperparameters + early-stopping budget of

10,000 rounds on the T8 validation set [42]), and a single-hidden-layer MLP (sklearn `MLPRegressor` architecture [43], implemented in PyTorch with AdamW and batch size 4096 for the 25M-row training set). All three treat each segment as an independent tabular row, so they do not use the graph topology. Both tree baselines exceed T8 on point accuracy (Table 5.9); the MLP, flat-tabular and without the GNN message passing, falls below T8.

Table 5.9.: Non-GNN baselines on the same 100-graph scenario-level held-out test set (3,163,500 nodes). The baselines use default-style configurations from each model's official documentation; XGBoost adds an early-stopping budget on the same scenario-level held-out validation set used by T8, and the MLP is a PyTorch implementation of the sklearn-default architecture with the documented optimisation and batch-size deviations of Section 5.11. XGBoost `best_iteration` reached the 10,000-round budget cap; the validation curve was effectively flat over the last 1,000 rounds and the result is reported at the budget cap rather than at a triggered early stop.

Model	R^2	MAE (veh/h)	RMSE (veh/h)
T8 (GNN, primary UQ base)	0.5957	3.957	7.118
Deep Ensemble (GNN, 5 members)	0.6841	3.485	6.293
Random Forest (cuML, default-style)	0.6612	3.263	6.516
XGBoost (default-style + early stop)	**0.7414**	**2.774**	**5.693**
MLP (sklearn-style architecture, PyTorch)	0.4928	3.883	7.973

Conformal on Random Forest. Split conformal on the RF point predictions (random 50/50 node-level partition, seed 42 – a marginal-coverage sanity check rather than a scenario-level held-out comparison) reaches near-nominal marginal coverage with tighter radii than T8 (Table 5.10), reflecting RF's smaller residuals rather than a stronger UQ method.

Scope. Tree baselines dominate point accuracy on this sparse target, as expected for tabular data with 88.7% zero-mass response [44, 45]. The per-prediction UQ pipeline evaluated here – MC Dropout, regression σ-scaling, the heteroscedastic and CQR heads of Trials 9–11 – is not directly applicable to the default tree baselines because those methods rely on architectural dropout, soft probabilistic outputs, and a differentiable backbone. The architecture was fixed to the PointNetTransfGAT family to keep the UQ evaluation directly comparable with the Natterer et al. [4] baseline, so the systematic UQ

Table 5.10.: Split conformal on top of the Random Forest, compared with T8 split conformal from Section 5.3. The RF split is at the *node* level (random 50/50 partition of the 3.16M-node test set, seed 42), unlike the T8 scenario-level split, so the RF conformal numbers are a marginal-coverage sanity check rather than a scenario-level held-out comparison; see the caveat above. Both methods achieve near-nominal marginal coverage on their respective evaluation halves; RF's tighter intervals reflect its smaller residuals, not a stronger uncertainty method.

Model	q_{90} (veh/h)	PICP_{90}	q_{95} (veh/h)	PICP_{95}
T8 (GNN) + split conformal	9.920	90.02%	14.677	95.01%
Random Forest + split conformal	**8.737**	89.98%	**12.964**	94.98%

comparison was deliberately scoped to GNN-based methods. Tree-native UQ extensions (quantile regression forests, NGBoost) would be the obvious follow-up to run; we do not test them and therefore make no claim about their relative quality.

5.12. An Illustrative Decision Framework

Figure 5.8 sketches one way to combine the chapter's findings into an accept/review/reject decision rule based on the predicted σ percentile. The framework is illustrative: thresholds, costs and risk tolerances would need to be calibrated against the planner's tolerances and the specific decision cost trade-offs before any real deployment. The stratified caveat from Section 5.10 also applies: uncertainty ranking weakens on the largest-$|\Delta v|$ segments, exactly the regime where policy effects are biggest; high-impact decisions should still go through MATSim regardless of σ.

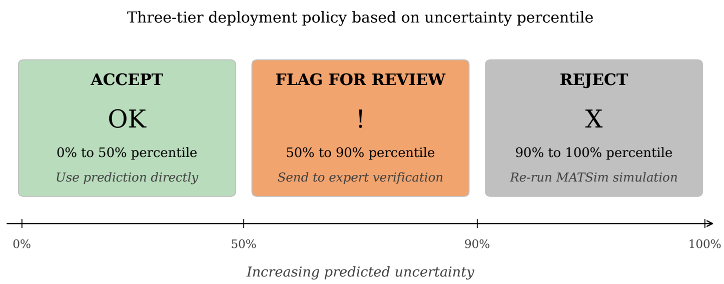

Figure 5.8.: Illustrative three-tier framework: bottom 50% accepted (−41.2% MAE, Fig. 5.3), middle 40% reviewed, top 10% rejected (AUROC = 0.7548, Section 5.5). Thresholds need calibration against the planner's cost trade-offs.

6. Discussion

6.1. Freezing the Backbone: An Empirical Observation

Trials 10 and 11 share head, loss, data, calibration protocol and learning rate (5×10^{-4}, inherited from T8 in both cases); the only design difference is whether the backbone is fine-tuned end-to-end (T10) or frozen (T11). T10 collapses to $R^2 = 0.4057$ and fails three of the six gates; T11 reaches $R^2 = 0.5835$ and passes all six (Tables 5.7, 5.8). T9 (frozen backbone, heteroscedastic head, NLL loss) reaches $k_{95} = 2.84$ – four times better than T8 – but misses the R^2 gate at 0.4991, consistent with the NLL/MSE trade-off of Section 2.3.3. For the architecture, subset and loss/head pairings tested here, the backbone-trainability flag alone is enough to flip the CQR result; we report this as a finding for the tested setup, not a universal one.

> *When adding an uncertainty-aware head on top of an MSE-trained backbone, freezing the backbone and training only the head appears to be the safer choice in the experimental setting studied here.*

6.2. The Layered Calibration Hierarchy

The four post-hoc treatments of σ compose into a layered pipeline (Table 6.1). MC Dropout supplies the σ ranking, σ-scaling gives a quick approximate 1σ probability, and adaptive conformal gives the deployment interval with near-nominal empirical marginal coverage on the held-out scenarios.

6.3. Stratified UQ: Why ρ Drops on Large $|\Delta v|$

The Q1→Q4 contrast ($\rho = 0.721 \rightarrow 0.100$) is best read as "MC Dropout works mechanically on trivial segments and breaks on hard ones" rather than as "MC Dropout is highly informative on easy regimes": Q1 is dominated by $|\Delta v| = 0$ segments where |error| and σ both reduce to functions of small output magnitudes, so the high Q1 ρ is partly an artefact of shared dependence. The substantive Q4 degradation is real and has three contributing mechanisms: (i) **dynamic range** – Q4 has $|\bar{y}| \approx 14$ veh/h and MAE

Table 6.1.: Layered calibration hierarchy: each row adds one capability. Numbers from Sections 5.2–5.5.

Layer	Ranking ρ	ECE	1σ cov.	90% marginal cov.	Conditional cov. range
Raw MC Dropout	0.4820	0.356	32.7%	48.55%	—
+ σ-scaling ($T = 2.887$)	0.4820	**0.034**	**68.0%**	—	—
+ Standard conformal	0.4820	—	—	**90.02%**	[59.0%, 98.1%]
+ Adaptive conformal	0.4820	—	—	89.87%	[**83.7%**, **96.4%**]

Each row reports the metric native to that calibration layer. σ-scaling targets the 1σ Gaussian probability and leaves the conformal residual quantile untouched; conformal prediction targets the marginal 90% interval and does not change the underlying σ.

10.08 vs Q1's $|\bar{y}| \approx 0$ and MAE 1.24, so absolute error spans a much wider range than absolute σ and Spearman noise dominates; (ii) **near-capacity saturation** – many Q4 segments sit near the knee of the flow–density curve where small policy changes trigger non-linear equilibrium shifts that MC Dropout variance does not capture (informed by traffic-equilibrium domain knowledge, not measured directly here); (iii) **sparse training coverage** – large-$|\Delta v|$ segments are rare in the 1,000-scenario subset. Mitigations include heteroscedastic heads (T9 captures aleatoric uncertainty, dominant here), explicit OOD detectors, and feature-conditional conformal methods.

6.4. The Exchangeability Assumption

Every conformal statement here rests on exchangeability, which is not strictly satisfied at the node level: nodes within a single graph share the same policy input and sit in a spatially correlated road network. The calibration and evaluation splits are therefore at the *scenario* level (50 calibration / 50 evaluation graphs, randomly assigned with seed 42). Exchangeability is enforced by splitting scenarios, and is more plausible at the scenario level than at the node level; node-level dependence remains within each graph and is not removed by the scenario split. The coverage guarantee should therefore be read as a marginal-over-scenarios statement: averaged across scenarios (and within a scenario, across nodes), at least 90% or 95% of residuals fall inside the conformal interval, as the observed 90.02% and 95.01% confirm. Within a single scenario, coverage can deviate; the per-graph standard-conformal coverage range is [80.5%, 93.7%] at the 90% target. A natural extension is graph-level nonconformity scoring (aggregating residuals to a per-graph score before computing the conformal quantile), which would restore strict exchangeability at the node-aggregate level at the cost of coarser intervals.

6.5. The PIT Diagnostic and CRPS/MAE Ratio

PIT mean 0.433 (ideal 0.500) and first-bin mass 28.4% indicate asymmetric underdispersion with an upward bias in $\hat{\mu}$. σ-scaling reduces the PIT KS from 0.245 to 0.104 but cannot shift the predictive centre, so conformal prediction handles the residual tails because its intervals expand around $\hat{\mu}$ regardless of where $\hat{\mu}$ sits; CQR (Trial 11) supports asymmetric intervals when the residual distribution itself is asymmetric. The CRPS/MAE ratio of 0.857 exceeds the Gaussian-calibrated optimum $1/\sqrt{2} \approx 0.707$ by $\approx 21\%$, quantifying the calibration cost of underdispersed σ.

6.6. Answers to the Research Questions

Each answer below is structured the same way: a direct answer, the supporting evidence, and the main limitation.

RQ1. How effectively does MC Dropout capture epistemic uncertainty? MC Dropout works as a *ranking* signal on Trial 8 but is not a calibrated probability model. Evidence: pooled $\rho = 0.4820$ between σ and $|\text{error}|$; mean per-graph $\rho = 0.464$ with bootstrap 95% CI $[0.460, 0.469]$; AUROC for top-10% error detection 0.7548; retaining the 50% most confident predictions reduces MAE by 41.2%. Limitation: $k_{95} = 11.66$ vs Gaussian 1.96, so raw σ is not a calibrated standard deviation without an additional scaling or conformal step (RQ4).

RQ2. How does MC Dropout compare with ensemble-based UQ? Deep Ensemble wins on point accuracy, MC Dropout on uncertainty ranking. Evidence: Deep Ensemble $R^2 = 0.6841$ at $\rho = 0.3997$, $k_{95} = 15.18$; MC Dropout $\rho = 0.4820$, $k_{95} = 11.66$. Limitation: only one architecture family tested; architecturally diverse ensembles could change this trade-off and were not evaluated.

RQ3. Does combining MC Dropout with seed-ensemble variance add benefit? Essentially no: quadrature gives $\rho = 0.4909$ vs 0.4908 for MC Dropout alone, within sampling noise. The useful signal in seed-ensemble variance is already captured by MC Dropout itself. Limitation: null result is specific to single-architecture ensembling; cross-family ensembles (GAT + GraphSAGE + GCN) might yield independent disagreement signals.

RQ4. Can distribution-free methods deliver trustworthy intervals without retraining? Yes, in an empirical marginal sense: the layered framework (MC Dropout + σ-scaling + adaptive conformal) requires no backbone retraining. Evidence: $\text{PICP}_{90} = 90.02\%$, $\text{PICP}_{95} = 95.01\%$ on T8; σ-scaling at $T = 2.887$ reduces ECE by 90.5% and hits Gaussian 1σ within 0.3 pp; adaptive conformal narrows conditional coverage from $[59.0\%, 98.1\%]$ to $[83.7\%, 96.4\%]$; Winkler scores $49.68 \rightarrow 35.78 \rightarrow 32.32$. Limitation: marginal coverage is averaged over scenarios. Exchangeability is enforced by the scenario-level split, but node-level dependence within a single graph is not removed (Section 6.4), so strong marginal UQ metrics do not by themselves imply policy-level reliability for any individual scenario.

RQ5. Can uncertainty-aware training extensions improve native UQ while preserving accuracy? Yes, but only when the MSE-trained backbone is kept frozen. Evidence: T9 (heteroscedastic, frozen) reaches $k_{95} = 2.84$ but misses the R^2 gate at 0.4991; T10 (CQR, full backbone unfrozen) collapses to $R^2 = 0.4057$ and fails three of six gates; T11 (same CQR head, frozen) passes all six gates at $R^2 = 0.5835$. The T10/T11 contrast isolates a single design knob – backbone trainability – and is the strongest empirical evidence for the freezing observation. Limitation: a setup-specific finding for PointNet-TransfGAT on the 1,000-scenario subset and the loss/head pairings tested; replication across architectures, datasets and backbone objectives is left for follow-up work.

6.7. Primary Findings

1. **MC Dropout works as a ranking signal at modest cost.** For T8, $\rho = 0.4820$ between σ and absolute error, with AUROC = 0.7548 for top-10% error detection. Across all 100 test scenarios no graph drops below $\rho = 0.41$.

2. **The calibration cascade closes the gap to near-nominal empirical marginal coverage.** Raw MC Dropout requires $k_{95} = 11.66$ for 95% coverage; regression σ-scaling at $T = 2.887$ brings this to 4.04 and reduces ECE by 90.5% (from 0.356 to 0.034). Adaptive conformal then narrows conditional coverage from $[59.0\%, 98.1\%]$ to $[83.7\%, 96.4\%]$ across uncertainty deciles.

3. **In these experiments, freezing the MSE-trained backbone is the strongest observed design choice when adding an uncertainty-aware head.** T9 (frozen, heteroscedastic head) is a partial positive; T10 (CQR, unfrozen backbone) collapses to $R^2 = 0.4057$; T11 (same CQR head, backbone frozen) clears all six gates with $R^2 = 0.5835$. With all other settings held fixed across T10 and T11, the contrast isolates the backbone-trainability flag. Broader claims would require replication across architectures, datasets, loss/head pairings, and backbone objectives.

4. **Deep ensembling and MC Dropout are complementary, not competing.** The five-member Deep Ensemble achieves the project's highest GNN-based, UQ-capable point accuracy ($R^2 = 0.6841$, +14.8% over T8) but its weakest UQ ranking ($\rho = 0.3997$). MC Dropout sits on the opposite corner of the trade-off. Pairing both at $\approx 5\times$ training compute gives the best of both.

5. **Stratified UQ has two regimes that should not be conflated.** On the smallest-$|\Delta v|$ quartile $\rho = 0.721$, but Q1 is dominated by segments with $|\Delta v| = 0$ (no policy effect at all), so the high correlation is partly a mechanical artefact. On the largest-$|\Delta v|$ quartile $\rho = 0.100$, a real breakdown driven by dynamic range, saturation effects, and sparse training-data coverage in the high-response regime. Uncertainty-guided filtering works best where the model is already good.

6.8. What the Existing Trials Do and Do Not Prove

The thesis runs many trials, but only a subset of design questions is directly answered by them; for the others, confounders remain. Table 6.2 summarises the main questions, the empirical answer reached here, the trial that supplies the evidence, and the confounds that remain. This is intended as a check on over-interpretation rather than a new experiment.

6.9. Limitations and Future Directions

- **Data scale.** 1,000 of 10,000 scenarios, reused across all eleven trials so that method-to-method comparisons sit on identical data. The resulting 3,163,500 node-level test predictions give per-graph Spearman ρ standard deviation 0.023 (bootstrap 95% CI ± 0.005 on the mean), which is the relevant precision for the relative UQ rankings that are the contribution here. The absolute $R^2 = 0.5957$ on T8 is below the $R^2 > 0.9$ reported by Natterer, Rao, Tejada Lapuerta, et al. [4] at full scale; replicating the UQ comparisons at full scale is the most direct piece of follow-up work.

- **T1 vs T8 confounds.** Final layer (Linear vs GATConv), batch size (32 vs 8) and learning rate (10^{-3} vs 5×10^{-4}) all differ. T1 has `use_dropout=False`, so MC Dropout is undefined; T8 is the strongest UQ-compatible base (Table 5.2).

- **Single network, single policy type.** Paris + capacity-reduction only; generalisation to other cities and policy types is untested.

- **Non-GNN baseline scope.** RF, XGBoost and MLP reported (Section 5.11); tree-native UQ extensions (quantile regression forests, NGBoost) not evaluated.

- **MC Dropout inference cost.** Inference scales linearly in the number of stochastic forward passes S; Trial 11 (frozen-backbone CQR) and GEBM [31] are single-forward-pass alternatives that eliminate this overhead when latency matters.

Table 6.2.: Summary of which design questions are answered by the existing trials, the answer in the present setting, and the remaining confounders that should be controlled in any follow-up.

Design question	**Answer**	**Evidence**	**Remaining confounders**
Does dropout hurt point accuracy here?	No (small)	T1 (no dropout, $R^2 = 0.786$) vs T2–T8 (with dropout, R^2 in 0.51–0.60); gap also reflects Linear vs GATConv head	Final-layer, batch size, and learning rate all differ between T1 and the dropout trials.
Does freezing the backbone help under CQR?	Yes, here	T10 (unfrozen, $R^2 = 0.4057$) vs T11 (same head, learning rate 5×10^{-4}, calibration; frozen, $R^2 = 0.5835$)	T10/T11 share head, loss, data, calibration, and LR. One untested alternative: the inherited LR (5×10^{-4}) may be too high for fine-tuning a converged backbone. Cross-architecture and cross-dataset replication left for future work.
Does ensembling improve point accuracy?	Yes, when independently seeded	5-member Deep Ensemble $R^2 = 0.6841$ vs single-member range $[0.640, 0.650]$; multi-model ensemble below best single trial	Deep Ensemble shares architecture and hyperparameters; architecturally diverse ensembles untested.
Does ensembling improve UQ ranking?	No, here	Deep Ensemble $\rho = 0.3997$ vs MC Dropout $\rho = 0.4820$; quadrature combination gives no gain over MC alone	Only one architecture family tested.
Does raw MC Dropout give calibrated intervals?	No	$k_{95} = 11.66$ vs Gaussian 1.96; Gaussian 95% interval covers only 54.85% of residuals	None major; follows from full-test-set audit.
Does regression σ-scaling fully calibrate?	No (rescales, can't reshape)	After $T = 2.887$, ECE drops 90.5% and 1σ hits 68.0%, but $2\sigma/3\sigma$ stay below Gaussian target	Calibration set is a node-level subset; node-within-graph correlation breaks strict scenario independence.
Does the freezing observation generalise?	Untested	Only PointNetTransfGAT on Paris 1,000-scenario subset under MSE-trained backbones is evaluated	Architecture, dataset, intervention type, and backbone objective all held fixed.

- **Future work.** Repeat at full 10,000-scenario scale; controlled T1-vs-T8 ablation; architecturally diverse ensembles (GAT + GraphSAGE + GCN); conformal risk control [19] and feature-conditional conformal; multi-city validation; explicit OOD detectors for the large-$|\Delta v|$ regime.

6.10. Threats to Validity

- **Dataset selection.** The 1,000-scenario subset is the first 20 contiguous batches of the 10,000-scenario Natterer corpus [4] rather than a uniform random sample, so a mild batch-ordering selection bias cannot be excluded. All eleven trials share the same subset, so the relative method-to-method comparisons that constitute the contribution remain valid; only the absolute accuracy and UQ figures inherit the potential bias.

- **External validity.** One road network (Paris), one intervention type (capacity reduction). Generalisation to other cities, topologies and policy types (road pricing, new infrastructure) is untested.

- **Model family.** All trials are PointNetTransfGAT variants. Transfer of the accuracy/UQ trade-offs and the freezing observation to other GNN families (GraphSAGE, GCN, message-passing transformers) or to non-graph models is untested.

- **Baseline scope.** RF, XGBoost and MLP reported (Section 5.11); tree-native UQ extensions (quantile regression forests, NGBoost) not evaluated, so whether tree-native methods can match MC Dropout's ranking or the adaptive conformal conditional-coverage range is left for follow-up work.

- **Metrics.** Strong UQ metrics (ρ, AUROC, PICP, k_{95}, CRPS) do not on their own imply policy-level usefulness. Whether uncertainty-guided filtering changes the quality of an actual planner decision requires user studies or simulated decision pipelines outside the scope here.

7. Conclusion

This thesis investigated uncertainty quantification for Graph Neural Network surrogates of agent-based transport models on a Paris road network. The work used a fixed 1,000-scenario subset of the 10,000-scenario Natterer corpus and evaluated eight base trials, three uncertainty-aware extensions, a five-member Deep Ensemble, six UQ families, and three post-hoc calibration and deployment tools on a 100-graph test set with 3,163,500 node-level predictions. The findings are specific to the Paris capacity-reduction setup studied here; the architecture (PointNetTransfGAT) and the scenario subset are held fixed throughout. The detailed research-question answers appear in Section 6.6, the primary findings in Section 6.7, and the threats to validity that bound these claims in Section 6.10.

Machine-learning surrogates can collapse a multi-hour MATSim run to seconds of GNN inference, but the speed advantage only translates into trustworthy decision support when the predictions come with a principled statement of reliability. The thesis showed that effective uncertainty quantification for GNN traffic surrogates is achievable at a tractable cost. MC Dropout supplies a ranking signal that holds up across scenarios; conformal prediction turns that signal into an empirical marginal coverage statement on the held-out evaluation graphs; and a frozen-backbone quantile head delivers calibrated intervals at a single deterministic forward pass. Compute costs differ across the toolkit – MC Dropout multiplies inference by a factor of S, the Deep Ensemble multiplies training by roughly $5\times$, and the frozen-backbone CQR head adds only 134 trainable parameters at deployment – yet all three are practical inside an offline policy-evaluation loop. MC Dropout and conformal prediction sit post-hoc on top of the trained backbone, while the CQR variant additionally trains a small new head.

A few ideas are worth keeping separate here. Ranking is not calibration, calibration is not coverage, and a backbone trained for one objective is not always a good starting point for a head trained under another. The freezing-backbone observation is a concrete example of that last point: in the experiments here, simply keeping the MSE-trained backbone frozen was enough to flip the CQR result from negative to positive. That makes it a sensible starting point for further work on uncertainty-aware extensions for this class of surrogates, with broader architectural and dataset coverage than was feasible inside the present scope.

To contextualise the GNN UQ stack against simpler tabular learners, three non-GNN baselines are reported in Section 5.11: Random Forest, XGBoost and a single-hidden-layer MLP, all on the same scenario-level held-out test set as T8 with default-style configurations from each model's official documentation. Both tree baselines exceed T8 on point accuracy (R^2 of 0.66 for Random Forest and 0.74 for XGBoost, against 0.60 for T8), in line with the tabular-learning literature for sparse-response targets. The focus of this thesis is on reliability around each prediction rather than peak point accuracy: a per-prediction uncertainty pipeline – MC Dropout, regression σ-scaling, conformal coverage and frozen-backbone CQR – that the default tree baselines do not provide, because the methods are tightly coupled to architectural dropout, soft probabilistic outputs, and a differentiable backbone with continuous representations. Tree-native UQ extensions such as quantile regression forests and NGBoost are the obvious next comparison to run; this work does not claim that no tree-based pipeline can match the GNN UQ stack, only that the default configurations tested here do not.

A. Verified Master Table

This appendix collects the numerical results of the thesis in one place, grouped by topic. Every value in the tables below has been recomputed from the saved per-trial prediction arrays and cross-checked against the canonical metric files produced during evaluation. As a reminder for cross-table comparisons: T1–T6 are evaluated on 50 test graphs (80/15/5 split, 1,581,750 nodes), while T7–T11 and the Deep Ensemble use 100 test graphs (80/10/10 split, 3,163,500 nodes). Comparisons of R^2 across the two groups should therefore be read descriptively, not as strict apples-to-apples.

A.1 Point Prediction Metrics (UQ-Capable Models)

Table A.1.: Point-prediction metrics for the UQ-capable models on the 100-graph test set (3,163,500 nodes); T1–T7 base trial metrics are in Section 5.1. T9 reports the MC mean; T10 and T11 report the quantile midpoint.

Model	R^2	MAE (veh/h)	RMSE (veh/h)
T8 (primary UQ base)	0.5957	3.957	7.118
T9 (MC mean)	0.4991	4.053	7.924
T10 (midpoint)	0.4057	4.130	8.631
T11 (midpoint)	0.5835	4.302	7.225
Deep Ensemble	**0.6841**	**3.485**	**6.293**

A.2 T8 MC Dropout UQ Metrics

Table A.2.: T8 MC Dropout, $S = 30$, full 3,163,500-node test set unless noted.

Metric	Value
Spearman ρ (σ vs $\lvert y - \hat{y} \rvert$)	0.4820
Mean per-graph ρ (mean / std)	0.464 / 0.023
Per-graph ρ range	[0.407, 0.505]
Bootstrap 95% CI for mean per-graph ρ ($B = 10{,}000$)	[0.460, 0.469]
Mean uncertainty $\bar{\sigma}$ (veh/h)	1.369
σ range (veh/h)	[0.042, 44.29]
k_{50}, k_{70}, k_{80}, k_{90}, k_{95}	1.71, 3.09, 4.55, 7.74, 11.66
Coverage at 1.96σ	54.85%
AUROC top-10% errors	0.7548
AUROC top-20% errors	0.7324
S-convergence: ρ at $S = 30$ / $S = 50$	0.466 / 0.471
S-convergence: gain $S = 30 \to 50$	+1.03%
T7 cross-check: Spearman ρ / k_{95} / AUROC top-10%	0.4437 / 16.15 / 0.7416

A.3 Conformal Prediction (T8)

Table A.3.: Standard split conformal and adaptive (σ-scaled) conformal on T8. Random scenario-level partition (seed 42, 50 calibration / 50 evaluation graphs, 1,581,750 nodes each).

Method	Level	Quantile q (veh/h)	PICP
Standard conformal	90%	9.920	90.02%
Standard conformal	95%	14.677	95.01%
Adaptive conformal	90% (global)	$q^{\mathrm{adapt}} = 7.71$	89.87% [1]
Conditional coverage range across deciles D1–D10:			
Standard conformal	90%	range	[59.0%, 98.1%]
Adaptive conformal	90%	range	[83.7%, 96.4%]

[1] Adaptive conformal global PICP_{90} = 89.87% is 0.13 pp below the 90% nominal target – within finite-sample tolerance but not strictly above the target.

A.4 Proper Scoring Rules (T8)

Table A.4.: CRPS, PIT and Winkler scores including PIT KS on the full 3,163,500-node T8 test set.

Metric	Value	Note
Mean CRPS (veh/h)	3.384	—
CRPS / MAE	0.857	Gaussian-calibrated optimum $1/\sqrt{2} \approx 0.707$
PIT mean	0.433	ideal 0.500
PIT KS statistic	0.245	—
PIT first-bin mass	28.4%	—
Winkler 90% (raw MC)	49.68	—
Winkler 90% (standard conformal)	35.78	−28.0% vs raw
Winkler 90% (adaptive conformal)	**32.32**	−35.0% vs raw

Abbreviations

ABM	Agent-Based Model
AUROC	Area Under the Receiver Operating Characteristic Curve
AUPRC	Area Under the Precision–Recall Curve
BNN	Bayesian Neural Network
CIT	Computation, Information and Technology (TUM school)
CQR	Conformalized Quantile Regression
CRPS	Continuous Ranked Probability Score
DE	Deep Ensemble
ECE	Expected Calibration Error
GAT	Graph Attention Network
GNN	Graph Neural Network
GPU	Graphics Processing Unit
KS	Kolmogorov–Smirnov (statistic)
MAE	Mean Absolute Error
MATSim	Multi-Agent Transport Simulation
MC	Monte Carlo
ML	Machine Learning
MLP	Multi-Layer Perceptron
MSE	Mean Squared Error
NLL	Negative Log-Likelihood
NPZ	NumPy compressed archive (file format)
OOD	Out-of-Distribution
PICP	Prediction Interval Coverage Probability
PIT	Probability Integral Transform
PyG	PyTorch Geometric
ReLU	Rectified Linear Unit
RMSE	Root Mean Squared Error
ROC	Receiver Operating Characteristic

RQ	Research Question
TUM	Technical University of Munich
UQ	Uncertainty Quantification

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